1	CHM.5.5.01.001.07 Perform interconversion between units of temperature and heat	Textbook + table 1 + Example problem 1 + Practice problems

How much energy in joules is supplied by a breakfast containing 170 Cal?	ما مقدار الطاقة بوحدة الجول (J) التي تزودنا بها وجبة افطار تحوي 170 Cal؟
170 J	○
7.11×10^5 J	○
1.7×10^5 J	○
711 J	○

Which of the following statements is true about the two figures in the table below?

أي العبارات التالية صحيحة فيما يتعلق بالشكلين في الجدول أدناه؟

 150 Cal	 5.0×10^5 J
Y	X

The energy in X is greater than in Y	الطاقة في X أكبر منها في Y
The energy in Y is 3.6×10^5 J	الطاقة في Y تُساوي 3.6×10^5 J
The energy in X is 355 Cal	الطاقة في X تُساوي 355 Cal
The energy in Y is greater than in X	الطاقة في Y أكبر منها في X

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Which of the following contains **the highest** nutritional Calories?

أي مما يلي يحتوي على **أكبر** كمية من السعرات الغذائية Cal؟

Learning Outcomes Covered

- CHM.5.5.01.001

a. 10 Cal

b. 9600 J

c. 86.5 kJ

d. 1000 cal

DR MOHAMED ABDELSALAM 0502500589

How much energy in joules is supplied by a breakfast bar containing 240 nutritional calories?

ما مقدار الطاقة بالجول التي توفرها وجبة الإفطار التي تحتوي على 240 Cal ؟

a. 240 J

b. 1.00×10^6 J

c. 5.74×10^4 J

d. 1004 J

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3	CHM.5.5.01.001.09 Describe how the same amount of heat affects the temperature of different objects of the same mass	Textbook + Example problem2 + Practice problems+ check your progress

A 355 g sample of unknown substance was heated from 22.4 °C to 43.6 °C the substance piece absorbs 6.75 kJ of energy.

Using the table below, which is the substance?

Substance	الذهب Gold	الفضة Silver	الألمنيوم Aluminum	الحديد Iron
Specific heat	0.129	0.235	0.897	0.449

a. Iron الحديد

b. Aluminum الألمنيوم

c. Silver الفضة

d. Gold الذهب

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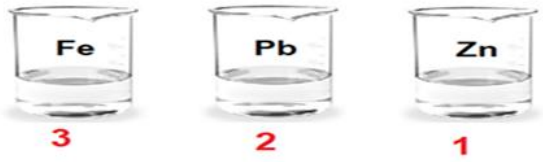
The following table shows specific heat of three metals:

الجدول التالي يوضح الحرارة النوعية لثلاثة فلزات:

Metal الفلز	Fe	Pb	Zn
specific heat الحرارة النوعية	0.449 J/(g.°C)	0.129 J/(g.°C)	0.388 J/(g.°C)

If **1.00 g** of each metal is heated to **100°C** and added to **10.0 g** of H_2O at **25°C** in the three separate glass beakers as in the figure below,

إذا سخن **1.00 g** من كل فلز إلى **100°C** وأُضيف إلى **10.0 g** من H_2O عند **25°C** في ثلاثة كؤوس زجاجية كما في الشكل أدناه،

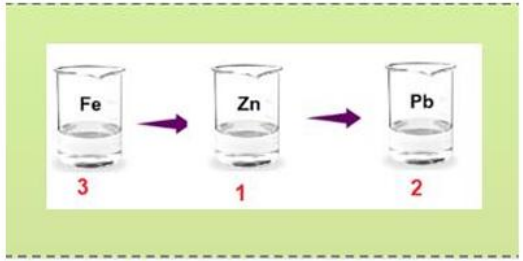


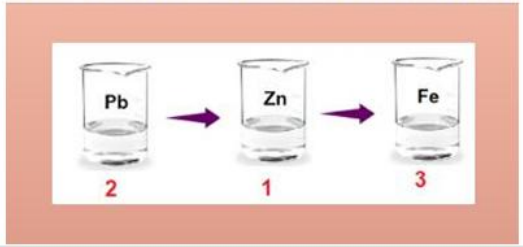
what is the order of the final temperatures of beakers from **the lowest to the highest?**

ما ترتيب درجات الحرارة النهائية للكؤوس من **الأقل إلى الأعلى؟**









DR MOHAMED ABDELSALAM 0502500589

Equal masses of of the four metals given in the table below were left to sit in the Sun at the same time and for the same length of time. What is the order of the four metals according to its temperature increase from smallest increase to largest?

متساوية من الفلزات الأربعة الواردة في الجدول أدناه في نفس الوقت ولنفس الفترة الزمنية. ما ترتيب الفلزات لزيادة درجة حرارتها من الأقل إلى الأعلى؟

الكالسيوم Calcium	السترونشيوم Strontium	الرصاص Lead	المغنيسيوم Magnesium	Metal الفلز
0.647	0.301	0.129	1.023	J /(g.^{o}C) Specific Heat الحرارة النوعية

Learning Outcomes Covered

- CHM.5.5.01.008

a. The lowest is calcium, then magnesium, then lead, then strontium

الأقل هو الكالسيوم ثم المغنيسيوم ثم الرصاص ثم السترنشيوم

b. The lowest is calcium, then strontium, then magnesium, then lead

الأقل هو الكالسيوم ثم السترنشيوم ثم المغنيسيوم ثم الرصاص

c. The lowest is magnesium, then calcium, then strontium, then lead

الأقل هو المغنيسيوم ثم الكالسيوم ثم السترنشيوم ثم الرصاص

d. The lowest is lead, then strontium, then magnesium, then calcium

الأقل هو الرصاص ثم السترنشيوم ثم المغنيسيوم ثم الكالسيوم

DR MOHAMED ABDELSALAM 0502500589

The same amount of heat is added to a **10 g** sample of each of the following metals. If each metal is initially at **20.0°C**, which metal will reach the highest temperature?

تضاف نفس كمية الحرارة إلى عينة كتلتها **10 g** من كل الفلزات التالية. إذا كانت درجة الحرارة الابتدائية لكل فلز هي **20.0°C** ما الفلز الذي يصل إلى أعلى درجة حرارة؟

The metal الفلز	specific Heat الحرارة النوعية
Beryllium البريليوم	1.82 J/(g.°C)
Calcium الكالسيوم	0.653 J/(g.°C)
Copper النحاس	0.385 J/(g.°C)
Gold الذهب	0.129 J/(g.°C)

a. Beryllium البريليوم

b. Calcium الكالسيوم

c. Copper النحاس

d. Gold الذهب

DR MOHAMED ABDELSALAM 0502500589

Equal masses of metals given in the following table were left to sit in the Sun at the same time and for the same length of time

تم ترك كتل متساوية من الفلزات الواردة في الجدول التالي في الشمس في نفس التوقيت ونفس المدة الزمنية

The metal الفلز	Aluminium الألمنيوم	Gold الذهب	Iron الحديد	Silver الفضة
Specific heat الحرارة النوعية	0.897 J/(g.°C)	0.129 J/(g.°C)	0.449 J/(g.°C)	0.235 J/(g.°C)

Which metal is the highest increasing in the temperature?

أي الفلزات له أعلى زيادة في درجة الحرارة؟

a. Aluminium الألمنيوم

b. Gold الذهب

c. Iron الحديد

d. Silver الفضة

DR MOHAMED ABDELSALAM 0502500589

2	CHM.5.5.01.002.02 Calculate the amount of heat released by a substance as its temperature changes	Textbook + Example problem2 + Practice problems
4	CHM.5.5.01.002.03 Calculate the specific heat capacity of a sample given its mass and temperature change	Textbook + Example problem 3 + Practice problems

When a 360 g piece of hot alloy is placed in 425 g of cold water in a calorimeter, the temperature of the alloy decreases by 205°C, while the temperature of the water increases by 18.7 °C.

What is the specific heat of the alloy?

Specific heat of water= 4.184 (J/g.°C)

a. 0.129 J/(g.°C)

b. 0.235 J/(g.°C)

c. 0.380 J/(g.°C)

d. 0.450 J/(g.°C)

DR M

time. What is the mass of the piece of iron (g) whose temperature increases by the same amount as the piece of aluminum?

الحديد Iron	الألمنيوم Aluminium	المادة Substance
.....	47.0 g	Mass الكتلة
0.449	0.897	الحرارة النوعية Specific Heat J/(g. ^{0}C)
30.0^0 C	30.0^0 C	ΔT

a. 93.9

b. 45.6

c. 35.5

d. 53.5

DR MOHAMED ABDELSALAM 0502500589

If 250 g of ethanol at 75.0°C loses 4655 J of heat, what is the final temperature of ethanol?
(Specific Heat of ethanol = 2.44 J/ (g. ^{0}C)

250 من الإيثانول درجة حرارته 75.0^{0}C كمية من الحرارة 4655 J . فما درجة الحرارة النهائية للإيثانول؟
النوعية للإيثانول هي (2.44 J/ (g. ^{0}C)

Learning Outcomes Covered

- CHM.5.5.01.008

a. 82.6^{0}C

b. 67.4^{0}C

c. 59.5^{0}C

d. 45.8^{0}C

A 15.0 g piece of cadmium metal absorbed 32.0 cal of heat. The final temperature was 62.7°C and the specific heat of cadmium equals 0.231 J/(g.°C).
What was the initial temperature?

امتصت قطعة كتلتها 15.0 g من فلز الكادميوم 32.0 cal من الحرارة، كانت درجة الحرارة النهائية 62.7°C والحرارة النوعية للكادميوم تساوي 0.231 J/(g.°C)
ماذا كانت درجة الحرارة الابتدائية؟

38.7°C

24.1°C

9.3°C

53.4°C

DR MOHAMED ABDELSALAM 0502500589

How much heat is absorbed by 3.00 kg granite boulder as temperature changes from 10.0°C to 45.0°C? (Specific heat of granite = 0.803 J/(g.°C)

ما كمية الحرارة التي تمتصها صخرة من الجرانيت كتلتها 3.00 kg عندما تتغير درجة حرارتها من 10.0°C إلى 45.0°C؟ (الحرارة النوعية للجرانيت = 0.803 J/(g.°C))

a. 8.4×10^4 J

b. 84 J

c. 1.1×10^5 J

d. 11 J

DR MOHAMED ABDELSALAM 0502500589

What is the amount of heat (J) absorbed when 125 g of aluminium is heated from 25.0 ^{0}C to 105.0 ^{0}C?

ما مقدار الحرارة (J) التي يتم امتصاصها عندما يتم تسخين 125 g من الألمنيوم من درجة حرارة 25.0 ^{0}C إلى 105.0 ^{0}C ؟

The specific heat of aluminium is 0.897J/(g. ^{0}C)	الحرارة النوعية للألمنيوم =(0.897 J/(g. ^{0}C

Learning Outcomes Covered

- CHM.5.5.01.008

a. 2690

b. 5860

c. 4650

d. 8970

DR MOHAMED ABDELSALAM 0502500589

Because you are interested in the heat given off by the chemical reaction going on inside the pack, it is convenient to think of the pack and its contents as the **system.** In thermochemistry, the system is the specific part of the universe that contains the reaction or process you wish to study. Everything in the universe other than the system is considered the **surroundings.** Therefore, the **universe** is defined as the system plus the surroundings.

universe = system + surroundings

What kind of energy transfer occurs during the exothermic heat-pack reaction? Heat produced by the reaction flows from the heat pack (the system) to your cold hands (part of the surroundings).

What happens in an endothermic reaction or process? The flow of heat is reversed. Heat flows from the surroundings to the system. When barium hydroxide and ammonium thiocyanate crystals, shown in **Figure 7,** are placed in a beaker and mixed, a highly endothermic reaction occurs. Placing the beaker on a wet board allows heat to flow from the water and board (the surroundings) into the beaker (the system). The temperature change is great enough that the beaker freezes to the board.

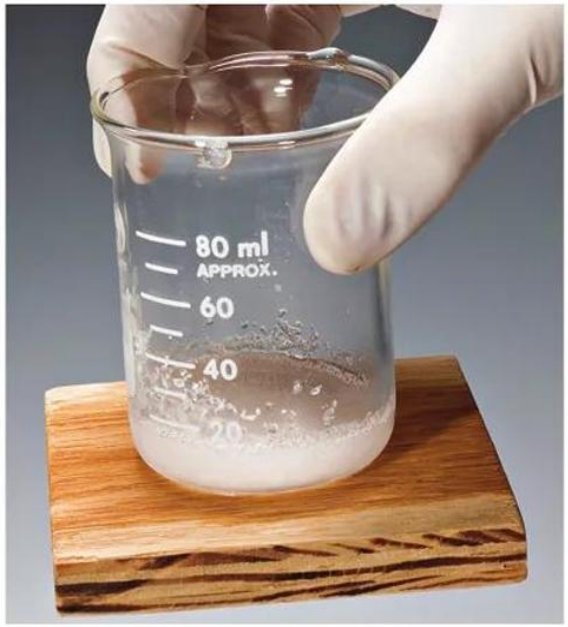


Figure 7 In this endothermic reaction, the reacting mixture draws enough energy from the water and the wet board to cause the wet board to freeze to the beaker.

According to the principles of thermochemistry described, the **universe** is defined as the combination of:

A. The reaction or process being studied only. B. The system plus the contents of the heat pack.
C. The system plus the surroundings. D. Everything that is *not* considered the surroundings.

DR MOHAMED ABDELSALAM 0502500589

6	CHM.5.5.01.004.03 Identify direction of heat flow (For Exothermic and endothermic processes)	Textbook + figures 8 ,9

In the endothermic reaction shown below,

What is the heat flows direction?

في التفاعل الماص للحرارة الموضح أدناه،

ما اتجاه انتقال الحرارة؟



Learning Outcomes Covered

- CHM.5.5.01.004

a. From the beaker to the wet board and water — من الكأس إلى اللوح الرطب والماء

b. From the mixture to the universe — من الخليط إلى الكون

c. From the wet board and water to the beaker — من اللوح الرطب والماء إلى الكأس

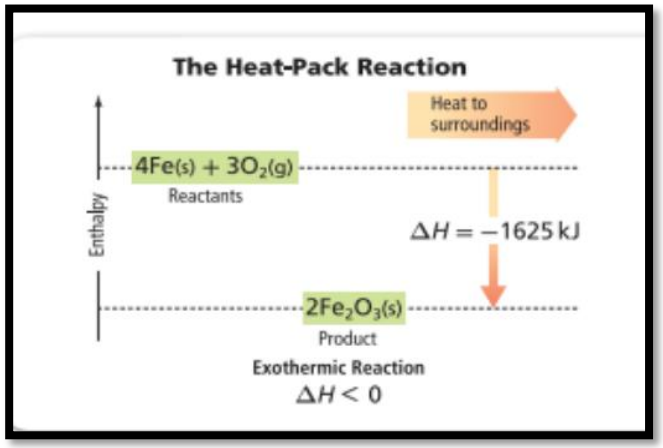
d. From the system to the surroundings — من النظام إلى المحيط

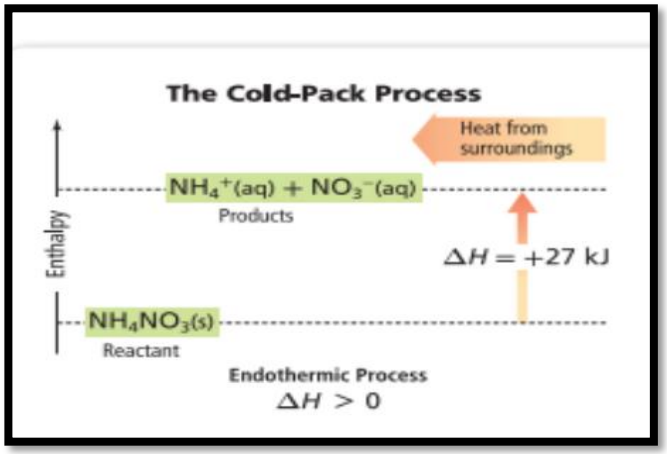
In the highly endothermic reaction described (barium hydroxide and ammonium thiocyanate crystals), the heat flow causes the wet board (part of the surroundings) to freeze to the beaker (the system). This indicates that, for this endothermic process:

A. The surroundings are gaining heat from the system. B. The system is releasing heat into the surroundings. C. The temperature of the system is increasing significantly. D. The system is absorbing heat *from* the surroundings.

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7	CHM.5.5.01.004.04 Compare and contrast potential energy diagrams of exothermic and endothermic reactions in terms of general shape, enthalpy of reactants and products, activation energy of forward and backward reactions, and enthalpy of reaction and its sign	Textbook + figures 8 ,9 + check your progress

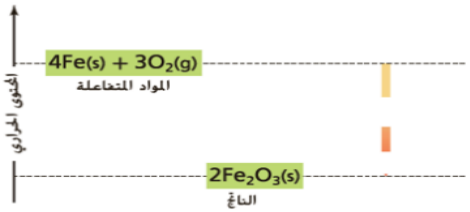




Which of the following is true regarding the figure below?

أي مما يأتي صحيح فيما يتعلق بالشكل أدناه؟





The sign of enthalpy change is positive	إشارة التغير في المحتوى الحراري موجبة
This reaction is used in the Cold–Pack	يُستخدم هذا التفاعل في الكمادة الباردة
The reaction is endothermic	التفاعل ماص للحرارة
This reaction is used in the Heat–Pack	يُستخدم هذا التفاعل في الكمادة الساخنة

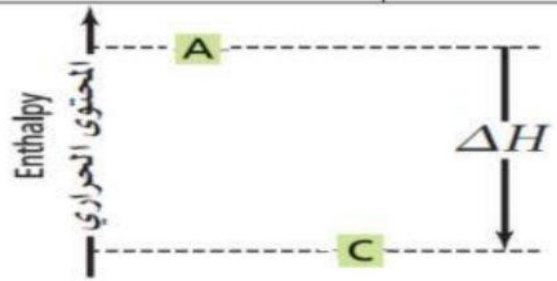
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Which of the following is **correct** about the reaction with the equation: $A \rightarrow C$,shown in the diagram below?

I.	$\Delta H > O$
II.	$H_{products} < H_{reactants}$
III.	Heat flows from the system to the surroundings

أي من التالية **صحيحًا** حول التفاعل ذو المعادلة: $A \rightarrow C$ الموضّح بالشكل أدناه؟

$\Delta H > O$	.I
$H_{\text{المتفاعلات}} < H_{\text{النواتج}}$	.II
الحرارة تنتقل من النظام إلى المحيط	.III



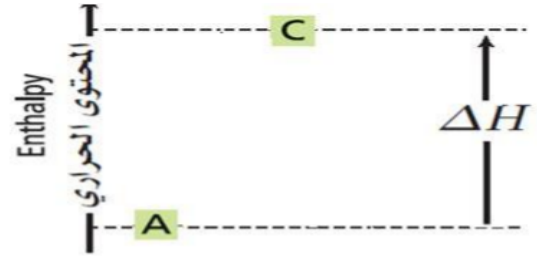
A. I only	A. I فقط
B. I and II only	B. I و II فقط
C. II and III only	C. II و III فقط
D. I and III only	D. I و III فقط

Which of the following is **correct** about the reaction with the equation: $A \rightarrow C$,shown in the diagram below?

I.	$\Delta H > O$
II.	$H_{products} < H_{reactants}$
III.	Heat flows from the surroundings to the system

أي من التالية **صحيحًا** حول التفاعل ذو المعادلة: $A \rightarrow C$ الموضّح بالشكل أدناه؟

$\Delta H > O$	.I
$H_{\text{المتفاعلات}} < H_{\text{النواتج}}$	.II
الحرارة تنتقل من المحيط الى النظام	.III

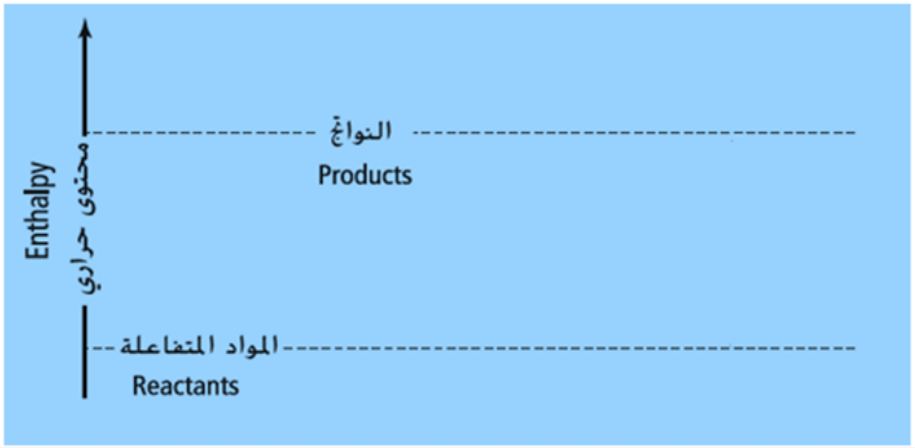


A. I only	A. I فقط
B. I and III only	B. I و III فقط
C. II and III only	C. II و III فقط
D. I and II only	D. I و II فقط

DF

Which of the following is **correct** regarding the figure below?

أي التالية **صحيح** فيما يتعلق بالشكل أدناه؟



The reaction is an exothermic	التفاعل طارد للحرارة
Heat to surroundings	الحرارة إلى الوسط
Represents the reaction that occur in the Cold-Pack	يُمثل التفاعل الذي يحدث في الكمادة الباردة
The enthalpy of the products is less than the enthalpy of the reactants	المحتوى الحراري للنواتج أقل من المحتوى الحراري للمواد المتفاعلة

DR MOHAMED ABDELSALAM 0502500589

Which of the following statements is incorrect About the heat pack?

العبارات التالية غير صحيحة حول الكمادة الساخنة؟



Learning Outcomes Covered

- CHM.5.5.02.004

a. Heat flows from heat pack to person cold hands

الحرار ناتجة من الكمادة الساخنة إلى يدي الشخص الباردتين

b. Heat flows from the surroundings to the system

الحرار ناتجة من المحيط إلى النظام

c. Heat pack is the system and person cold hands is surroundings

ة السا هي النظام ويدي الشخص الباردتين هي المحيط

d. The reaction in this pack is exothermic

ل في الكمادة طارد للحرارة

Regarding the heat –pack ,which of the following is correct?

فيما يتعلق بالكمادة الساخنة، أي مما يلي صحيح ؟





a. $\Delta H > 0$ $\Delta H > 0$

b. $H_{products} > H_{reactants}$ $H_{\text{نواتج}} > H_{\text{متفاعلات}}$

c. Heat from surrounding الحرارة من الوسط المحيط

d. The sign of ΔH is negative إشارة ΔH سالبة

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$$4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s) \quad \Delta H = -1625 \text{ kJ}$$

$$NH_4NO_3(s) \rightarrow NH_4^+(aq) + NO_3^-(aq) \quad \Delta H = 27 \text{ kJ}$$

$$H_2O(l) \rightarrow H_2O(g) \qquad \Delta H_{vap} = 40.7 \text{ kJ}$$

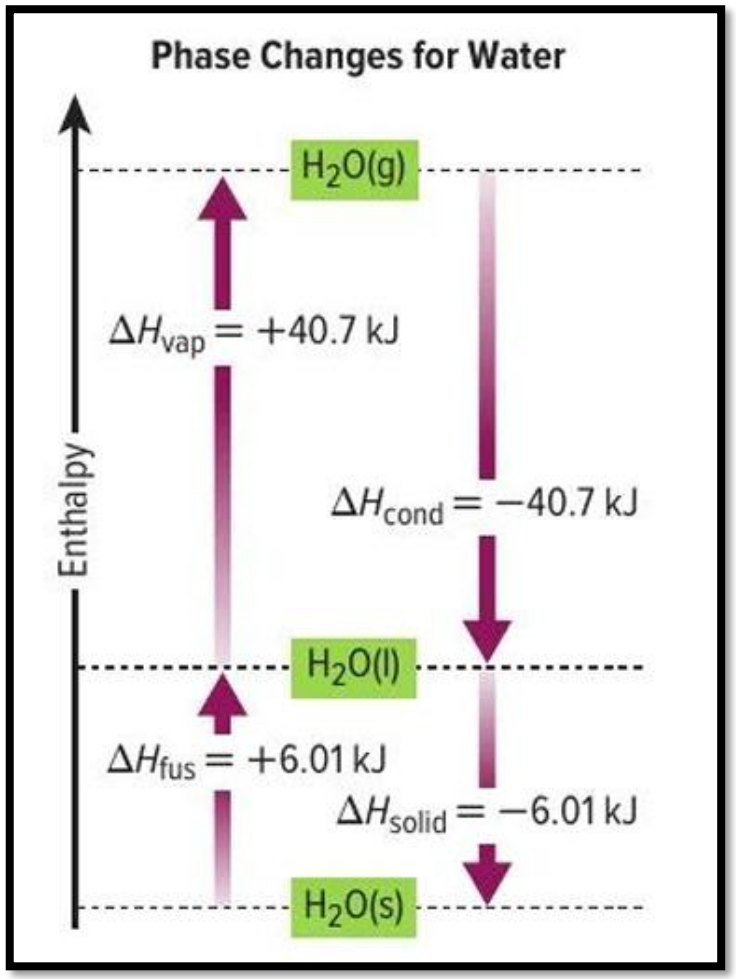
$$H_2O(s) \rightarrow H_2O(l) \qquad \Delta H_{fus} = 6.01 \text{ kJ}$$

$$H_2O(g) \rightarrow H_2O(l) \qquad \Delta H_{cond} = -40.7 \text{ kJ}$$

$$H_2O(l) \rightarrow H_2O(s) \qquad \Delta H_{solid} = -6.01 \text{ kJ}$$

$$\Delta H_{vap} = -\Delta H_{cond}$$

$$\Delta H_{fus} = -\Delta H_{solid}$$



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Which of the following represents the cold pack process?

أي مما يلي يُمثل العملية التي تحدث في الكمادة الباردة؟



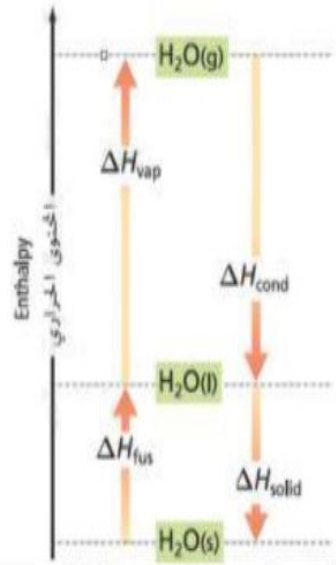
a. $NH_4NO_{3(s)} \rightarrow NH^+_{4(aq)} + NO^-_{3(aq)} \qquad \Delta H_{rxn} < 0$

b. $NH_4NO_{3(s)} + 27\ kJ \rightarrow NH^+_{4(aq)} + NO^-_{3(aq)}$

c. $NH_4NO_{3(s)} + 27\ kJ \rightarrow NH^+_{4(aq)} + NO^-_{3(aq)} + 27\ kJ$

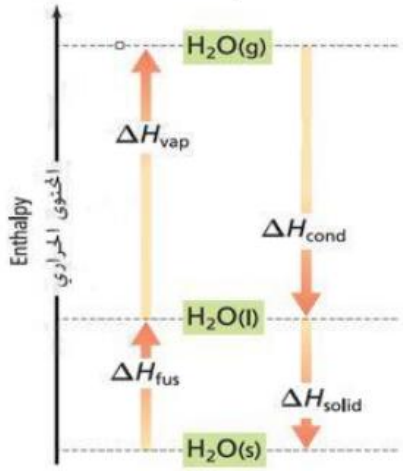
d. $NH_4NO_{3(s)} \rightarrow NH^+_{4(aq)} + NO^-_{3(aq)} \qquad \Delta H_{rxn} = -27\ kJ$

DR MOHAMED ABDELSALAM 0502500589

Which of the following statements is not correct according to the diagram shown below?	أي العبارات التالية غير صحيحة اعتمادًا على الشكل أدناه؟
	
A. The ΔH values of the molar enthalpy of vaporization and the molar enthalpy of fusion are positive	A. تكون قيم ΔH للحرارة المولية للتبخير والحرارة المولية للانصهار موجبة
B. The ΔH values of the molar enthalpy of condensation and molar enthalpy of solidification are negative	B. تكون قيم ΔH للحرارة المولية للتكثيف والحرارة المولية للتجمد سالبة
C. The molar enthalpy of solidification and the molar enthalpy of fusion have the same numerical values but with opposite signs	C. تتساوى القيمة العددية للحرارة المولية للتجمد مع القيمة العددية للحرارة المولية للانصهار ولكن تختلف إشارتهما
D. The molar enthalpy of condensation and the molar enthalpy of vaporization have the same numerical values with same sign	D. تتساوى القيمة العددية للحرارة المولية للتكثف مع القيمة العددية للحرارة المولية للتبخير وتتشابه إشارتهما

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Which of the following statement is correct according to the diagram shown below?	أي العبارات التالية صحيحة اعتمادًا على الشكل أدناه؟

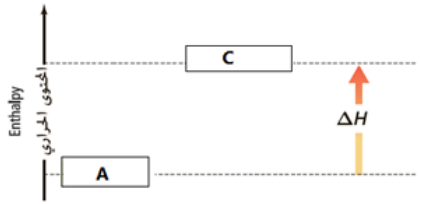


A. The system energy decreases as ice melts and then evaporates	A. تقلّ طاقة النظام عند انصهار الثلج ثم تبخره
B. The system energy increases as water vapor condenses and then solidifies	B. تزيد طاقة النظام عند تكثف بخار الماء ثم تجمّده
C. The vaporization of water and melting of ice are considered as exothermic process	C. تُعتبر عمليات تبخّر الماء وصهر الثلج عمليات طاردة للحرارة
D. The condensation and solidification of water are considered as exothermic process	D. تُعتبر عمليات تكثّف الماء وتجمده طاردة للحرارة

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Regarding the diagram below for the chemical reaction **A→C** that happened in pack, which of the following is correct ?

فيما يتعلق بالرسم البياني أدناه للتفاعل الكيميائي **A→C** الذي يحدث في كمامة، أي مما يلي صحيح ؟



	نوع الكمادة Type of pack	التفسير Explanation
1	ساخنة Hot	لأن المحتوى الحراري للمواد الناتجة (C) أكبر من المحتوى الحراري للمواد المتفاعلة (A) Because enthalpy of products (C) more than enthalpy of reactants(A)
2	ساخنة Hot	لأن $\Delta H < 0$ Because $\Delta H < 0$
3	باردة Cold	لأن المحتوى الحراري للمواد الناتجة (C) أقل من المحتوى الحراري للمواد المتفاعلة (A) Because enthalpy of products (C) less than enthalpy of reactants(A)
4	باردة Cold	لأن $\Delta H > 0$ Because $\Delta H > 0$

a. 1

b. 2

c. 3

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Which of the following processes is exothermic?

أي العمليات التالية تكون طاردة للحرارة؟

I	تبخير الماء *The vaporization of water*
II	انصهار الثلج *The melting of ice*
III	تكثف بخار الماء *The condensation of water vapor*
IV	تجمد الماء *The freezing of water*

a. I and II II

b. I and III III

c. II and IV IV و

d. III and IV IV و

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Which of the following processes are endothermic?

أي العمليات التالية ماصة للحرارة؟

$C_2H_5OH(l) \longrightarrow C_2H_5OH(g)$	1
$NH_3(g) \longrightarrow NH_3(l)$	2
$NaCl(s) \longrightarrow NaCl(l)$	3
$C_5H_{12}(l) + 8O_2(g) \longrightarrow 5CO_2(g) + 6H_2O(l)$	4

a. 1 and 2 — 2و1

b. 2 and 3 — 3و2

c. 1 and 3 — 3و1

d. 2 and 4 — 4و2

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9	CHM.5.5.01.003.05 Perform phase-change calculations using enthalpy of fusion, vaporization, condensation and solidification	fiqure 10 + Practice problems + check your progress

Using the figure below, how much heat in (kJ) is required to evaporate 156.1 g of water at 100°C?

(The molar mass of water is 18.02 g/mol)

ـ أدناه، ما كمية الحرارة بوحدة (kJ) اللازمة لتبخير من الماء عند درجة حرارة 100^0C؟

ية للماء تساوي 18.02 g/mol)

Learning Outcomes Covered

- CHM.5.5.01.003

a. 353

b. 236

c. 185

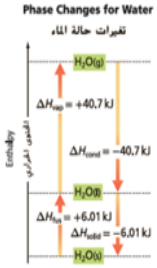
d. 347

What is the heat released in freezing 4.50 g water?

(Molar mass of water=18.02 g/mol)

ما مقدار الحرارة الناتجة عند تجمد 4.50 g من الماء؟

(الكتلة المولية للماء = 18.02 g/mol)



a. 1.50 kJ

b. 27.04 kJ

c. 10.16 kJ

d. 183.15 kJ

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Water is sprayed on oranges during a frosty night. If an average of **29.5 g** of water freezes on each orange, how much heat is released?

يتم رش الماء على البرتقال خلال الليالي الباردة. إذا كان متوسط مقدار الماء الذي يتجمد على كل برتقالة هو **29.5 g** فما كمية الحرارة الناتجة؟

(الكتلة المولية للماء = 18.02 g/mol)

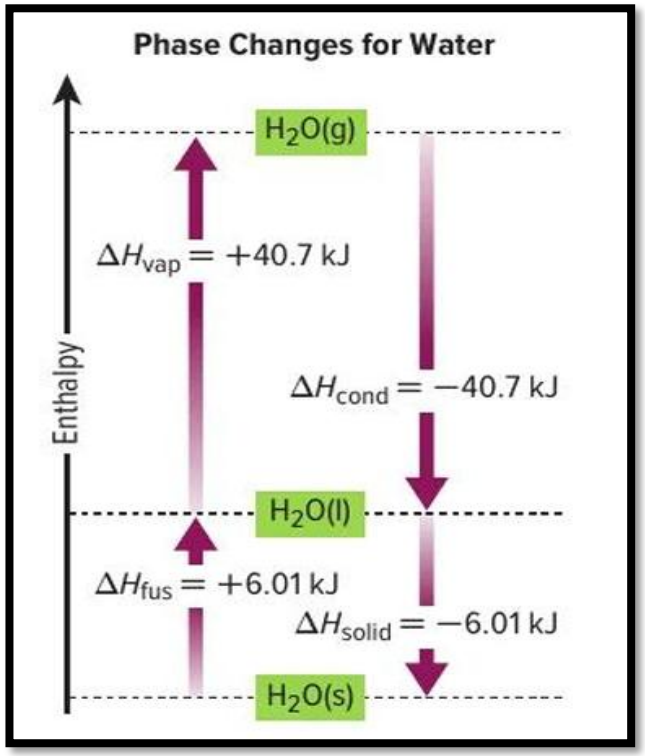
(molar mass of water = 18.02 g/mol)

a. 9.84 kJ

b. 66.6 kJ

c. 3195 kJ

d. 21636 kJ



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10	CHM.5.5.01.003.04 Define molar enthalpy (heat) of solidification, ΔHsolid	Textbook

The **molar enthalpy of solidification** (ΔH_{solid}) is defined as the amount of **heat energy released** when **one mole** of a substance changes phase from a **liquid to a solid** at its constant freezing point and a constant pressure.

The molar enthalpy of solidification is equal in magnitude but opposite in sign to the molar enthalpy of fusion (ΔH_{fus}), which is the heat absorbed when one mole of a solid melts to a liquid:

$$\Delta H_{solid} = -\Delta H_{fus}$$

$$H_2O(l) \rightarrow H_2O(s) \qquad \Delta H_{solid} = -6.01 \text{ kJ}$$

Some farmers make use of the heat of fusion of water to protect fruit and vegetables from freezing. If the temperature is predicted to drop to freezing, they flood their orchards or fields with water. When the water freezes, energy (ΔH_{fus}) is released and often warms the surrounding air enough to prevent frost damage.

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What mass of propane (C_3H_8) in unit (g) is burned to release 9985.5 kJ of heat?

ما كتلة البروبان (C_3H_8) بوحدة (g) التي يتم حرقها لإنتاج 9985.5 kJ من الحرارة؟

$$C_3H_{8\,(g)} + 5O_{2\,(g)} \longrightarrow 3CO_{2(g)} + 4H_2O_{(g)}$$

The enthalpy (heat) of combustion of propane $\Delta H^0_{comb} = -2219$ kJ/mol

(حرارة احتراق البروبان ΔH^0_{comb} هي -2219 kJ/mol والكتلة المولية للبروبان 44.097 g/mol)

Learning Outcomes Covered

- CHM.5.5.01.006

a. 198.0

b. 154.4

c. 110.3

d. 66.20

Using the equation below, how much heat ΔH_{comb} is released by burning 154 g of CH_4 methane?

(Molar mass of methane is 16.04 g/mol)

عادلة أدناه، ما كمية الحرارة الناتجة ΔH_{comb} عن احتراق ن الميثان CH_4 ؟

(الكتلة المولية للميثان هي 16.04 g/mol)

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + 891\ kJ$$

Learning Outcomes Covered

- CHM.5.5.01.007

a. 4.50×10^3 kJ

b. 5.20×10^3 kJ

c. 6.70×10^3 kJ

d. 8.60×10^3 kJ

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Enthalpy (heat) of Combustion: the change in enthalpy upon complete combustion of 1 mol of a substance.

What is ΔH_{comb} represents? → heat of combustion

1 mol + O_2 → ..... $\Delta H = \Delta H_{comb}$

element or compound | any mole number

combustion of 1 mol of methane produces 891 kJ of heat energy

$CH_{4(s)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + 891\ kJ$

combustion of 1 mol of octane produces 5471 kJ of heat energy

$C_8H_{18(s)} + \frac{25}{2}O_{2(g)} \rightarrow 8CO_{2(g)} + 9H_2O_{(l)} + 5471\ kJ$

combustion of glucose produces big amount of energy

$C_6H_{12}O_{6(s)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)}$ $\Delta H_{comb} = -2808\ kJ$

hydrogen and oxygen react together to provide the energy needed to lift the space shuttle $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)} + 286\ kJ$

Which of the following equations represents a molar combustion equation?

a. $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$

b. $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$

c. $C_6H_{12}O_{6(s)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(g)}$

d. $2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$

What is the complete thermochemical equation for the combustion of methane CH_4 if you know that $\Delta H_{comb} = -891$ kJ/mol?

a. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + 891\ kJ$

b. $CH_{4(g)} + 2O_{2(g)} + 891\ kJ \rightarrow CO_{2(g)} + 2H_2O_{(l)}$

c. $CO_{2(g)} + 2H_2O_{(l)} \rightarrow CH_{4(g)} + 2O_{2(g)} + 891\ kJ$

d. $CO_{2(g)} + 2H_2O_{(l)} + 891\ kJ \rightarrow CH_{4(g)} + 2O_{2(g)}$

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Which of the following equations does not represent combustion reaction?

أي المعادلات التالية لا تُمثل تفاعل احتراق؟

a. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} + 891\ kJ$

b. $C_8H_{18(l)} + \frac{25}{2}O_{2(g)} \rightarrow 8CO_{2(g)} + 9H_2O_{(l)} + 5471\ kJ$

c. $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)} + 286\ kJ$

d. $27\ kJ + NH_4NO_{3(s)} \rightarrow NH^+_{4(aq)} + NO^-_{3(aq)}$

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What mass of sucrose $C_{12}H_{22}O_{11}$ must be burned in order to liberate **8466 kJ** of heat?

ما كتلة السكروز $C_{12}H_{22}O_{11}$ التي يجب حرقها لإنتاج **8466 kJ** من الحرارة؟

Substance المادة	Formula الصيغة	ΔH°_{comb}	Molar mass الكتلة المولية
Sucrose السكروز	$C_{12}H_{22}O_{11}$	$-5644\ kJ/mol$	**342.3** g/mol

513.5 g

228.2 g

684.6 g

171.1 g

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How much heat is released by the combustion of 50 g of hydrogen gas?

(molar mass of H_2=2.0 g/mol)

ما كمية الحرارة الناتجة عن احتراق 50 g من غاز الهيدروجين؟

(الكتلة المولية لـ H_2 =2.0 g/mol)

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) + 286\ kJ$$

a. 29458 kJ

b. 7150 kJ

c. 14300 kJ

d. 572 kJ

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12	CHM.5.5.02.002.01 Calculate, using Hess's law, the ΔH of a reaction	Textbook + Example problem 5 + Practice problems

Using thermochemical equations, I,II and III below,
What is the $\Delta H°$ value for the following reaction?

مستخدمًا المعادلات الكيميائية الحرارية I و II و III أدناه ،
ما قيمة $\Delta H°$ للتفاعل التالي؟

$$ClF_{(g)} + F_{2(g)} \rightarrow ClF_{3(g)} \qquad \Delta H^{\circ} = ?$$

I	$2OF_{2(g)} \rightarrow O_{2(g)} + 2F_{2(g)}$	$\Delta H^{\circ} = -49.9$ kJ
II	$2ClF_{(g)} + O_{2(g)} \rightarrow Cl_2O_{(g)} + OF_{2(g)}$	$\Delta H^{\circ} = +205.6$ kJ
III	$ClF_{3(g)} + O_{2(g)} \rightarrow \frac{1}{2}Cl_2O_{(g)} + \frac{3}{2}OF_{2(g)}$	$\Delta H^{\circ} = +266.7$ kJ

Learning Outcomes Covered

- CHM.5.5.02.002

a. +422 kJ

b. +394 kJ

c. −139 kJ

d. −188 kJ

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How much is ΔH of the following reaction? $CO_{(g)} + 2H_{2(g)} \rightarrow CH_3OH_{(l)}$ Use the thermochemical equations (a, b and c) shown below	ما قيمة ΔH للتفاعل التالي؟ $CO_{(g)} + 2H_{2(g)} \rightarrow CH_3OH_{(l)}$ استخدم المعادلات الكيميائية الحرارية a، b و c الموضحة أدناه
a. $CO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -284$ kJ b. $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -286$ kJ c. $CH_3OH_{(l)} + \frac{3}{2}O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$ $\Delta H = -727$ kJ	

-1297 kJ

+1051 kJ

-129 kJ

+157 kJ

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What is the value of **ΔH** for the following reaction?

ما قيمة **ΔH** للتفاعل التالي؟

$$2H_2S_{(g)} + 3\,O_{2(g)} \rightarrow 2\,H_2O_{(l)} + 2SO_{2(g)}$$

Use thermochemical equations (a,b and c) shown below:

استخدم المعادلات الكيميائية الحرارية(a ، b و c) الموضحة أدناه:

a. $S_{(g)} + O_{2(g)} \rightarrow SO_{2(g)}$ $\Delta H = -296.8\ kJ$

b. $H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8\ kJ$

c. $H_{2(g)} + S_{(g)} \rightarrow H_2S_{(g)}$ $\Delta H = -20.6\ kJ$

−1124 kJ

−603.2 kJ

−1206 kJ

−562.0 kJ

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Using Hess's law and the changes in enthalpy for the following reactions

مُستخدمًا قانون هس والتغيرات في المحتوى الحراري للتفاعلات التالية

$$2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2(g)} \qquad \Delta H = -566.0\ kJ$$

$$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)} \qquad \Delta H = -180.6\ kJ$$

What is ΔH for the following reaction?

ما قيمة ΔH للتفاعل التالي؟

$$\mathbf{2CO_{(g)} + 2NO_{(g)} \rightarrow 2CO_{2(g)} + N_{2(g)}} \qquad \mathbf{\Delta H = ??}$$

a. −746.6 kJ

b. −466.0 kJ

c. −192.7 kJ

d. −385.4 kJ

DR MC

14	CHM.5.5.02.001.01 State Hess's Law - CHM.5.5.02.001.02 Describe the rules to be followed to apply Hess's Law	Textbook

Unfortunately, laboratory experiments to produce sulfur trioxide and determine its ΔH result in a mixture of products that is mostly sulfur dioxide (SO_2). In situations such as this, you can calculate ΔH by using Hess's law of heat summation. **Hess's law** states that if you can add two or more thermochemical equations to produce a final equation for a reaction, then the sum of the enthalpy changes for the individual reactions is the enthalpy change for the final reaction.

Chemists use a theoretical way to determine ΔH when:

1- The reaction is too slow.

2- The reaction occurs under conditions difficult to duplicate in a laboratory.

3- The reaction produces products other than the desired ones.

$$2S(s) + 3O_2(g) \rightarrow 2SO_3(g) \quad \Delta H = ?$$

Step 1 Chemical equations are needed that contain the substances found in the desired equation and have known enthalpy changes. The following equations contain the substances S, O_2, and SO_3.

a. $S(s) + O_2(g) \rightarrow SO_2(g) \quad \Delta H = -297$ kJ

b. $2SO_3(g) \rightarrow 2SO_2(g) + O_2(g) \quad \Delta H = 198$ kJ

Step 2 The desired equation shows two moles of sulfur reacting, so rewrite Equation **a** for two moles of sulfur by multiplying the coefficients by two. Double the enthalpy change ΔH because twice the energy will be released if two moles of sulfur react. With these changes, Equation **a** becomes the following (Equation **c**).

c. $2S(s) + 2O_2(g) \rightarrow 2SO_2(g) \quad \Delta H = 2(-297 \text{ kJ}) = -594$ kJ

Step 3 In the desired equation, sulfur trioxide is a product rather than a reactant, so reverse Equation **b.** When you reverse an equation, you must also change the sign of its ΔH. Equation **b** then becomes Equation **d.**

d. $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g) \quad \Delta H = -198$ kJ

Step 4 Add Equations **c** and **d** to obtain the desired reaction. Add the corresponding ΔH values. Cancel any terms that are common to both sides of the combined equation.

$$\begin{array}{ll} 2S(s) + 2O_2(g) \rightarrow 2SO_2(g) & \Delta H = -594 \text{ kJ} \\ 2SO_2(g) + O_2(g) \rightarrow 2SO_3(g) & \Delta H = -198 \text{ kJ} \\ \hline 2\cancel{SO_2}(g) + 2S(s) + 3O_2(g) \rightarrow 2\cancel{SO_2}(g) + 2SO_3(g) & \Delta H = -792 \text{ kJ} \end{array}$$

If you can add two or more thermochemical equations to produce a final equation for a reaction, then the sum of the enthalpy changes for the individual reactions is the enthalpy change for the final reaction.

1- Hess law

2- Le Chatelier's Principle

3- Gipps law

4- Henry law

500589

13	CHM.5.5.02.006.03 Calculate, using the standard enthalpies of formation, the Enthalpy of a reaction, ΔHrxn	Textbook + Example problem 6 + Practice problems

المادة Substance	ΔH_f° (kJ/mol)
$NO_{2(g)}$	33.2
$H_2O_{(l)}$	-285.8
$HNO_{3(aq)}$	-207.4
$NO_{(g)}$	91.3

What is the ΔH_{rxn}° value for the following reaction?

ما قيمة ΔH_{rxn}° للتفاعل التالي؟

$$3NO_{2(g)} + H_2O_{(l)} \rightarrow 2HNO_{3(aq)} + NO_{(g)} \quad \Delta H_{rxn}^\circ = ?$$

Learning Outcomes Covered

- CHM.5.4.02.006
- CHM.5.4.02.010
- CHM.5.4.02.021
- CHM.5.5.02.006

a. −137 kJ

b. -506 kJ

c. +136 kJ

d. + 368 kJ

DR M

Using the values of standard enthalpies(heat) of formation below, what is the ΔH°_{rxn} value for the following reaction?

مُستخدمًا قيم حرارة التكوين القياسية أدناه، ما قيمة ΔH°_{rxn} للتفاعل التالي؟

$$C_6H_{12}O_6(s) + 6\,O_2(g) \rightarrow 6\,CO_2(g) + 6\,H_2O(l)$$

$C_6H_{12}O_{6(s)}$	$H_2O_{(l)}$	$CO_{2(g)}$	substance المادة
-1273.3	-285.8	-393.5	(kJ/mol) ΔH°_f

a. −1901 kJ/mol

b. −2803 kJ/mol

c. −594 kJ/mol

d. −5349 kJ/mol

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Using standard enthalpies of formation below,
what is the ΔH°_{rxn} value for the following reaction ?

مُستخدمًا قيم حرارة التكوين القياسية أدناه ،
ما قيمة ΔH°_{rxn} للتفاعل التالي؟

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)}$$

$SO_{2(g)}$	$ZnO_{(s)}$	$ZnS_{(s)}$	substance المادة
-296.8	-348.3	-206.0	(kJ/mol) ΔH°_{f}

a. −901.3 kJ

b. −270.6 kJ

c. −593.7 kJ

d. −878.2 kJ

DR MOHAMED ABDELSALAM 0502500589

15	CHM.5.5.02.006.02 Identify the standard enthalpy (heat) of formation of a free element in its standard state	Textbook+ fiqure15

Standard heat of formation: the change in enthalpy that accompanies the formation of one mole of a compound from its elements that are in their standard state.

What is ΔH_f° represents? → standard heat of formation

element + element → 1 mol compound $\Delta H = \Delta H_f^\circ$

any mole number | any mole number

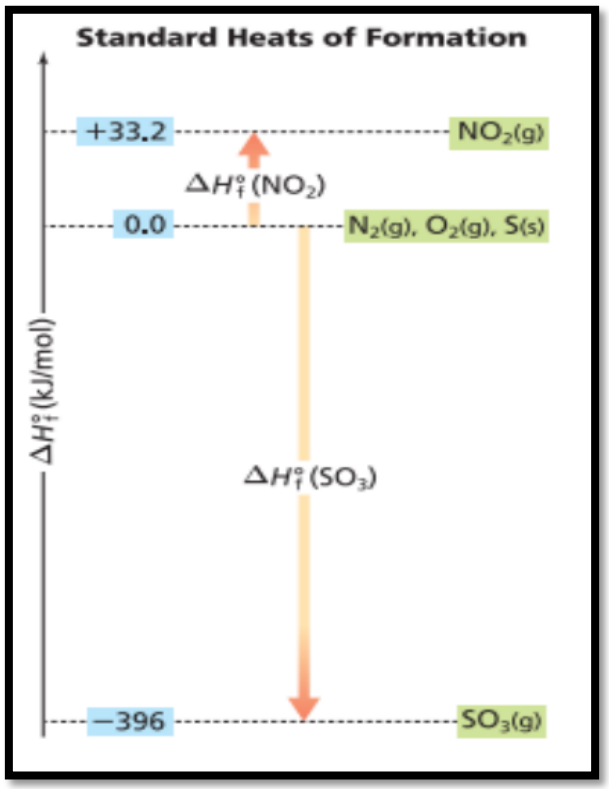


Table 5 Standard Enthalpies of Formation

Compound	Formation Equation	ΔH_f° (kJ/mol)
$H_2S(g)$	$H_2(g) + S(s) \rightarrow H_2S(g)$	−21
$HF(g)$	$\frac{1}{2}H_2(g) + \frac{1}{2}F_2(g) \rightarrow HF(g)$	−273
$SO_3(g)$	$S(s) + \frac{3}{2}O_2(g) \rightarrow SO_3(g)$	−396
$SF_6(g)$	$S(s) + 3F_2(g) \rightarrow SF_6(g)$	−1220

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Which of the enthalpy changes in the following reactions does not represent a standard heat of formation ΔH_f° ?

أي من التغيرات في المحتوى الحراري في التفاعلات التالية لا يُمثل حرارة تكوين قياسية ΔH_f° ؟

Learning Outcomes Covered

- CHM.5.5.02.006

a. $\frac{1}{2}H_{2(g)} + \frac{1}{2}F_{2(g)} \rightarrow HF_{(g)} \qquad \Delta H_f^\circ = -273\ kJ$

b. $S_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow SO_{3(g)} \qquad \Delta H_f^\circ = -396\ kJ$

c. $NO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{2(g)} \qquad \Delta H_f^\circ = -58.1\ kJ$

d. $S_{(s)} + 3F_{2(g)} \rightarrow SF_{6(g)} \qquad \Delta H_f^\circ = -1220\ kJ$

Which of the enthalpy changes in the following reactions represents a standard heat of formation (ΔH_f°) ?

تغيرات في المحتوى الحراري في التفاعلات التالية يُمثل حرارة سية (ΔH_f°) ؟

Learning Outcomes Covered

- CHM.5.5.02.006

a. $\frac{1}{2}N_{2(g)} + O_{2(g)} \longrightarrow NO_{2(g)}\ ,\ \Delta H = +33.2\ kJ$

b. $CO_{(g)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{2(g)}\ ,\ \Delta H = -283\ kJ$

c. $2S_{(s)} + 3O_{2(g)} \longrightarrow 2SO_{3(g)}\ ,\ \Delta H = -792\ kJ$

d. $2Fe_2O_{3(s)} \longrightarrow 4Fe_{(s)} + 3O_{2(g)}\ ,\ \Delta H = 1625\ kJ$

DR MOHAMED ABDELSALAM 0502500589

Which of the ΔH values given in the following reactions represents **the standard enthalpy of formation** ΔH_f° of NO_2 compound?

أي من قيم ΔH المُعطاة في التفاعلات التالية تُمثل **حرارة التكوين القياسية** ΔH_f° للمركب NO_2 ؟

1	$NO_{(g)} + O_{3(g)} \rightarrow NO_{2(g)} + O_{2(g)}$	$\Delta H = -199\ kJ$
2	$\frac{1}{2}N_{2(g)} + O_{2(g)} \rightarrow NO_{2(g)}$	$\Delta H = +33.2\ kJ$
3	$N_{2(g)} + 2O_{2(g)} \rightarrow 2NO_{2(g)}$	$\Delta H = +66.4\ kJ$
4	$NO_{(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{2(g)}$	$\Delta H = -58.1\ kJ$

What is the standard enthalpy (heat) of formation ΔH_f° for SO_3 compound?

ما حرارة التكوين القياسية ΔH_f° للمركب SO_3 ؟

$$2S(s) + 3O_2(g) \rightarrow 2SO_3(g) \quad \Delta H = -792 \text{ kJ}$$

a. −792 kJ

b. −396 kJ

c. +1584 kJ

d. +792 kJ

DR MOHAMED ABDELSALAM 0502500589

Which of the following equations represents a thermochemical equation for the formation of 1 mol of SO_3 from its elements in their standard states?

أي المعادلات التالية تُمثل المعادلة الكيميائية الحرارية لتكوين واحد مول من SO_3 من عناصره في حالتها القياسية؟

Learning Outcomes Covered

- CHM.5.5.02.006

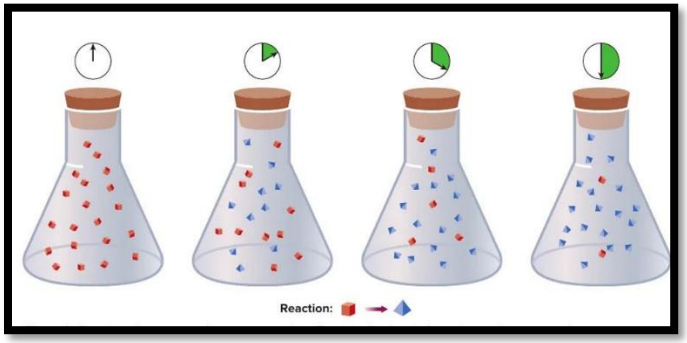
a. $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)} \qquad \Delta H^\circ = -198\ kJ$

b. $S_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow SO_{3(g)} \qquad \Delta H^\circ = -396\ kJ$

c. $3S_{(s)} + \frac{9}{2}O_{2(g)} \rightarrow 3SO_{3(g)} \qquad \Delta H^\circ = -1188\ kJ$

d. $2S_{(s)} + 3O_{2(g)} \rightarrow 2SO_{3(g)} \qquad \Delta H^\circ = -792\ kJ$

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Average Reaction Rate Equation

$$\text{Average reaction rate} = -\frac{\Delta[\text{reactant}]}{\Delta t}$$

quantity		unit
symbol	meaning	
Rate	reaction rate	mol/(L.s)
k	specific rate constant	different units
[A]	reactant concentration	mol/L or M

In the gas-phase reaction : $I_2 + Cl_2 \rightarrow 2ICl$

$[I_2]$ changes from 0.400M at 0.00 min to 0.3840 M at 4.00 min.

What is the average reaction rate in moles of I_2 consumed per liter per minute?

في التفاعل الغازي : $I_2 + Cl_2 \rightarrow 2ICl$

يتغير $[I_2]$ من 0.400 M عند 0.00 min إلى 0.3840 M في 4.00 min

ما متوسط سرعة التفاعل بمولات I_2 المستهلكة باللتر لكل دقيقة؟

+0.004 mol/(L.min)

− 0.004 mol/(L.min)

+0.01 mol/(L.min)

− 0.01 mol/(L.min)

Using the experimental data in the following table, if the average reaction rate for the reaction, expressed in moles of NO formed, is 0.050 mol/L.s , what concentration of NO would be present after 0.100 s?

مُستخدمًا البيانات التجريبية في الجدول التالي، إذا كان متوسط سرعة التفاعل مُعبرًا عنه بعدد مولات NO الناتجة يساوي 0.05 mol/L.s ما تركيز NO بعد 0.100 s؟

$N_2(g) + O_2(g) \rightarrow 2NO(g)$

$[NO]$ (M)	$[O_2]$ (M)	$[N_2]$ (M)	Time (s) الزمن
0.000	0.300	0.500	0.00
	0.250	0.450	0.100

a. 0.005

b. 0.5

c. 0.2

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In the gas – phase reaction $I_2 + Cl_2 \longrightarrow 2ICl$

$[I_2]$changes from 0.400 M at 0.00 s

to 0.300 M at 4.00 s.

What is the average reaction rate over

the given time period expressed in mol/L.s?

في التفاعل الغازي $I_2 + Cl_2 \longrightarrow 2ICl$

يتغير تركيز اليود I_2 من 0.400 M عند 0.00 s

إلى 0.300 M في 4.00 s .

ما متوسط سرعة التفاعل خلال الفترة الزمنية المعلومة بوحدة mol/L.s؟

a. 0.0250

b. 0.0350

c. 0.870

d. 0.690

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17	CHM.5.4.01.002.03 List the two conditions, according to the collision theory, that must be met for a collision between reactant molecules to be effective in producing new chemical species	Textbook + figures 3 , 4

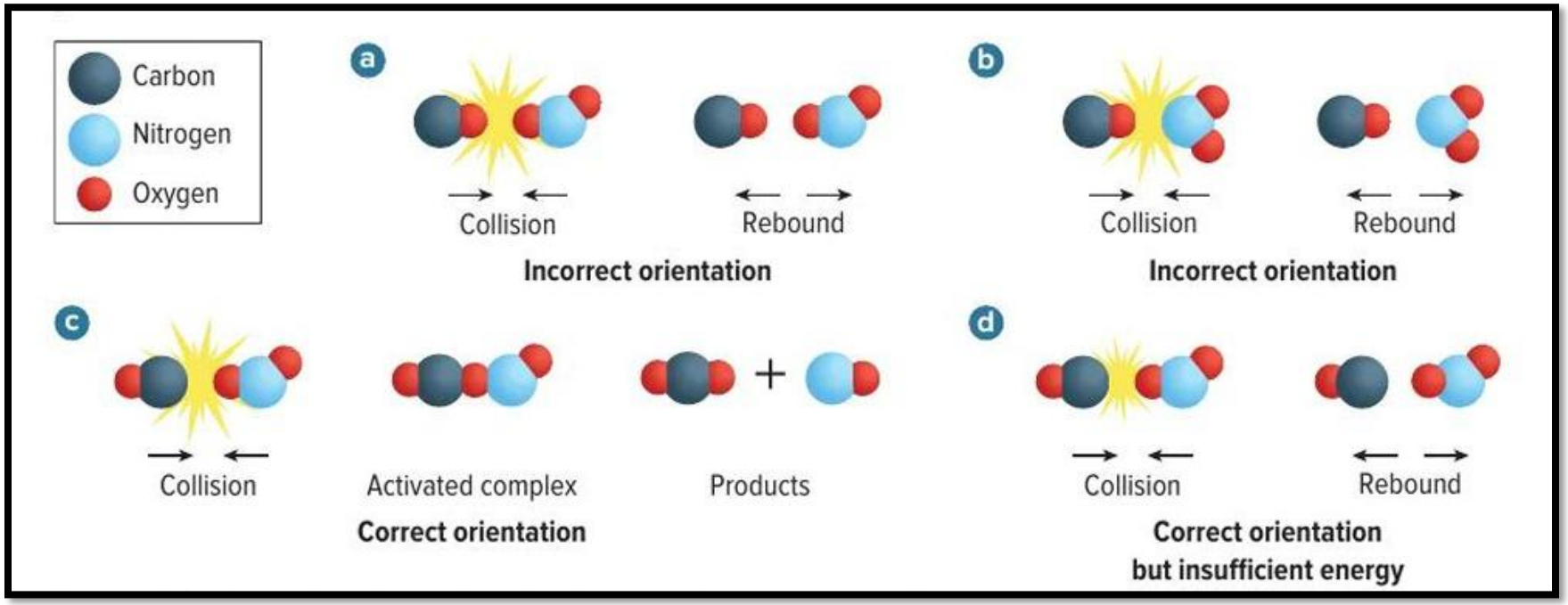


Table 1 Collision Theory Summary

1. Reacting substances (atoms, ions or molecules) must collide.
2. Reacting substances must collide in the correct orientation.
3. Reacting substances must collide with sufficient energy to form an activated complex.

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18	CHM.5.4.01.022.03 Describe the relationship between activation energy and rate of the reaction	Textbook + figure 5, 6

The higher the activation energy, the slower the rate of reaction, and conversely, the lower the activation energy, the faster the rate of reaction.

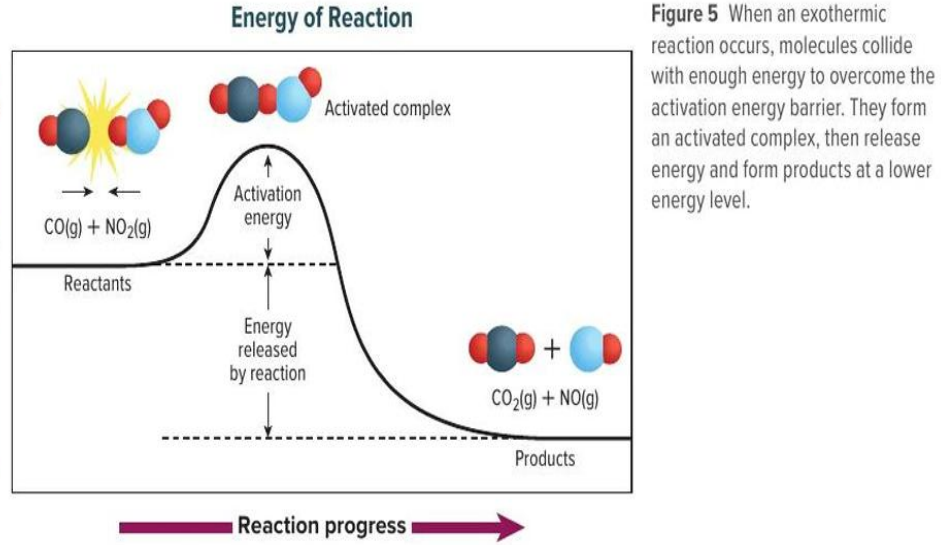


Figure 5 When an exothermic reaction occurs, molecules collide with enough energy to overcome the activation energy barrier. They form an activated complex, then release energy and form products at a lower energy level.

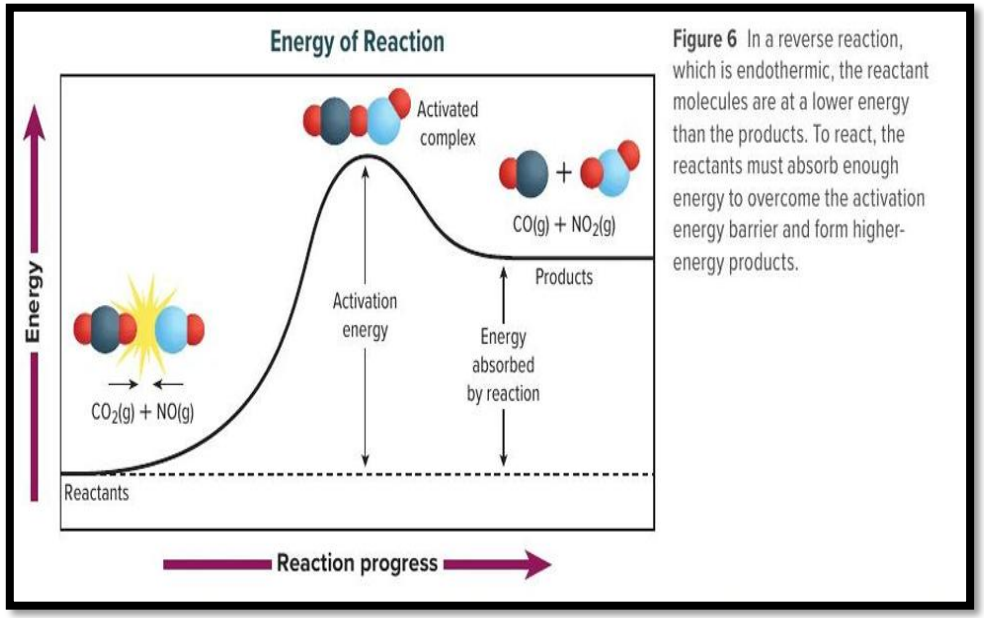


Figure 6 In a reverse reaction, which is endothermic, the reactant molecules are at a lower energy than the products. To react, the reactants must absorb enough energy to overcome the activation energy barrier and form higher-energy products.

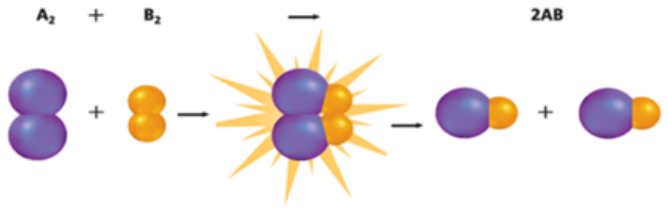
What is the relationship between activation energy and chemical reaction rate ?

1- Direct proportional
2- Inverse
3- The same
4- No relation

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Why the collisions between the reactant molecules success to produce products in the following reaction?

لماذا نجحت التصادمات بين جزيئات المواد المتفاعلة في إنتاج النواتج في التفاعل التالي؟



a. Because the molecules collide in the correct orientation, and they have sufficient energy to react

لأن الجزيئات تتصادم في الاتجاه صحيح ، ولديها الطاقة الكافية للتفاعل

b. Because the molecules collide in the incorrect orientation, and they have insufficient energy to react

لأن الجزيئات تتصادم في الاتجاه الخاطئ ، ولديها طاقة غير كافية للتفاعل

c. Because the molecules collide in the correct orientation, and they have insufficient energy to react

لأن الجزيئات تتصادم في الاتجاه صحيح ، ولديها طاقة غير كافية للتفاعل

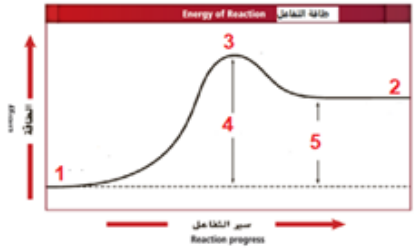
d. Because the molecules collide in the incorrect orientation, and they have sufficient energy to react

لأن الجزيئات تتصادم في الاتجاه الخاطئ ، ولديها الطاقة الكافية للتفاعل

DR MOHAMED ABDELSALAM 0502500589

Regarding the figure below, which of the following is correct?

فيما يتعلق بالشكل أدناه ، أي مما يلي صحيح ؟



a. 1 indicates products while 2 indicates reactants

يدل الرقم 1 على النواتج بينما يدل الرقم 2 على المواد المتفاعلة

b. 4 indicates activation energy while 5 indicates energy absorbed by reaction

يدل الرقم 4 على طاقة التنشيط بينما يدل الرقم 5 على الطاقة التي يمتصها التفاعل

c. 4 indicates activation energy while 5 indicates energy released by reaction

يدل الرقم 4 على طاقة التنشيط بينما يدل الرقم 5 على الطاقة الناتجة عن التفاعل

d. 1 indicates activated complex while 2 indicates reactants

يدل الرقم 1على المعقد المنشط بينما يدل الرقم 2على المواد المتفاعلة

DR MOHAMED ABDELSALAM 0502500589

Which of the following is **correct** regarding the two figures(1 and 2) below?

أي مما يلي **صحيح** فيما يتعلق بالشكلين(1و2) أدناه؟

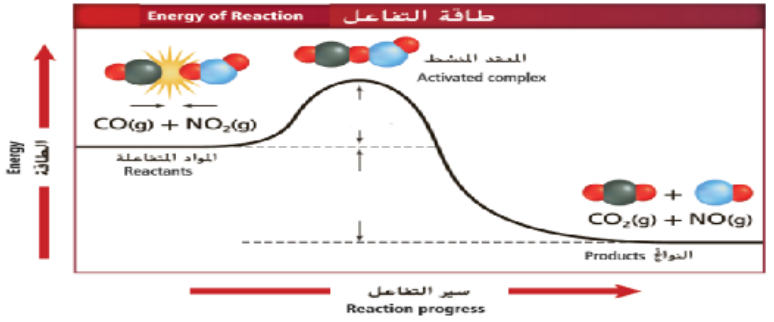


Figure 1 الشكل

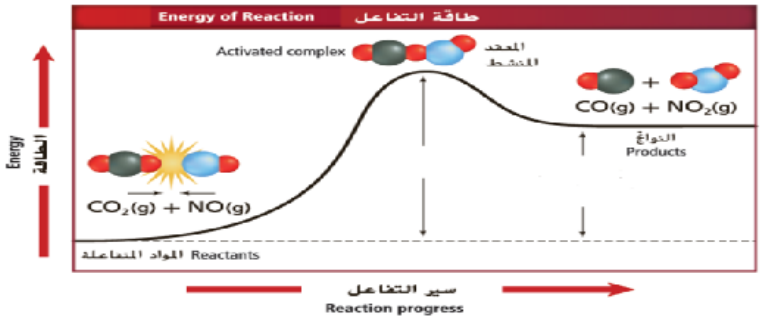


Figure 2 الشكل

In figure 1 the reactant molecules collide with enough energy to overcome the activation energy barrier and form **higher** energy products

في الشكل 1 تتصادم جزيئات المتفاعلات بطاقة كافية للتغلب على حاجز طاقة التنشيط وتكون النواتج عند مستوى طاقة **أكبر**

The activation energy in figure 1 **larger** than activation energy in figure 2

طاقة التنشيط في الشكل 1 **أكبر** من طاقة التنشيط في الشكل 2

The reaction in figure 2 is exothermic, while the reaction in figure1 it is endothermic

التفاعل في الشكل 2 طارد للحرارة، بينما التفاعل في الشكل 1 ماص للحرارة

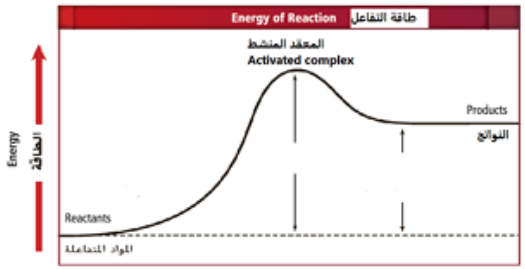
In figure 2 the reactant molecules absorb enough energy to overcome the activation energy barrier and form **higher** energy products

في الشكل 2 تمتص جزيئات المتفاعلات طاقة كافية للتغلب على حاجز طاقة التنشيط وتشكل نواتج ذات طاقة **أكبر**

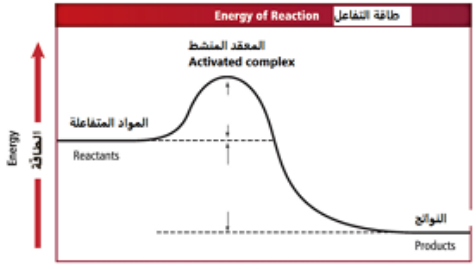
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Using the two figures below, which of the following is **correct**?

مُستخدمًا الشكلين أدناه، أي مما يلي **صحيح**؟



الشكل (2)
figure (2)



الشكل (1)
figure (1)

a. Energy is released in figure (1) while energy is absorbed in figure (2)

تنطلق طاقة في الشكل (1) بينما تُمتص طاقة في الشكل (2)

b. Energy is released in figure (2) while energy is absorbed in figure (1)

تنطلق طاقة في الشكل (2) بينما تُمتص طاقة في الشكل (1)

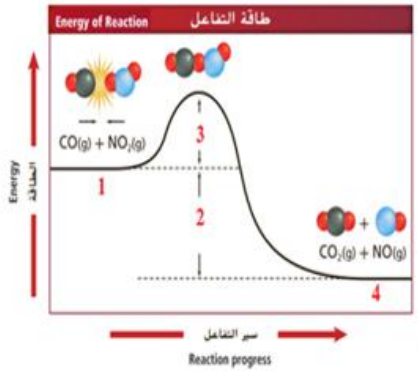
c. The activation energy in figure (1) is greater than the activation energy in figure (2)

طاقة التنشيط في الشكل (1) أكبر من طاقة التنشيط في الشكل (2)

\

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represented by the graph below?



4	3	2	1	
النواتج Products	طاقة التنشيط Activation energy	الطاقة الناتجة عن التفاعل Energy released by reaction	المواد المتفاعلة Reactants	A
المواد المتفاعلة Reactants	الطاقة الناتجة عن التفاعل Energy released by reaction	طاقة التنشيط Activation energy	النواتج Products	B
المواد المتفاعلة Reactants	طاقة التنشيط Activation energy	النواتج Products	الطاقة الناتجة عن التفاعل Energy released by reaction	C
الطاقة الناتجة عن التفاعل Energy released by reaction	النواتج Products	المواد المتفاعلة Reactants	طاقة التنشيط Activation energy	D

a. A
b. B
c. C
d. D

DR MOHAMED ABDELSALAM 0502500589

19	CHM.5.4.01.005.05 Explain how the exponents in the rate equation for a chemical reaction relate to the coefficients in the chemical equation	Textbook

$$aA + bB \rightarrow \text{products}$$

The general rate law for such reactions is described below.

The General Rate Law

$$\text{rate} = k[A]^m[B]^n$$

[A] and [B] represent the concentrations of reactants A and B. The exponents *m* and *n* are the reaction orders.

The rate of a reaction is equal to the product of *k* and the concentrations of the reactants each raised to a power (order) that is determined experimentally.

Only if the reaction between A and B occurs in a single step (and with a single activated complex) does $m = a$ and $n = b$. That is unlikely, however, because single-step reactions are not common. For example, consider the multi-step reaction between nitrogen monoxide (NO) and hydrogen (H_2), which is described by the following equation.

$$2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$$

This reaction has the following experimentally determined rate law.

$$\text{rate} = k[NO]^2[H_2]$$

If the next reaction occurs in a single step and with a single activated complex

The rate law for the reaction $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

So, m= 2 n =

When: m = a, n = b?	When the reaction between A and B occurs in one step and with only one activated complex.
Does the reaction: aA + bB → products always occur in one step?	No, because one-step reactions are not common.

One-step Reaction: A → B	Reaction Rate: Rate = k [A]
The reaction: aA + bB → products	Reaction Rate: Rate = $k[A]^m[B]^n$

What is the rate law for the reaction $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ that is first order in O_2 and third order in terms of reaction order?

a. Rate = $k\,[NO]^2[O_2]$

b. Rate = $k\,[NO]^3[O_2]$

c. Rate = $k\,[NO]^2[O_2]^3$

d. Rate = $k\,[NO][O_2]$

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20	CHM.5.4.01.006.02 Use the method of initial rates to write rate law of reaction	Textbook + Practice problems

Use the experimental data in the following table.

What is the rate law for the reaction?

$$aA + bB \rightarrow C$$

استخدم البيانات التجريبية في الجدول التالي .

ما قانون السرعة للتفاعل؟

$$aA + bB \rightarrow C$$

Experimental Data البيانات التجريبية

التجربة Trial	التركيز الابتدائي Initial concentration $[A]$ (M)	التركيز الابتدائي Initial concentration $[B]$ (M)	السرعة الابتدائية Initial Rate mol/(L.s)
1	1.2	2.4	8×10^{-8}
2	1.2	1.2	4×10^{-8}
3	3.6	2.4	7.2×10^{-7}

$$\text{Rate} = k[A]^2$$

$$\text{Rate} = k[A][B]$$

$$\text{Rate} = k[A]^2[B]$$

$$\text{Rate} = k[A][B]^2$$

Using the experimental data in the following table, what is the rate law for the reaction?

مُستخدمًا البيانات التجريبية في الجدول التالي، ما قانون السرعة للتفاعل؟

$aA + bB \rightarrow products$			
التجربة Trial	التركيز الابتدائي Initial concentration $[A]$ (M)	التركيز الابتدائي Initial concentration $[B]$ (M)	السرعة الابتدائية Initial Rate (mol/(L.s))
1	0.273	0.273	3.00×10^{-3}
2	0.546	0.273	3.00×10^{-3}
3	0.546	0.546	6.00×10^{-3}

a. $\text{Rate} = k[A]$

b. $\text{Rate} = k[A][B]$

c. $\text{Rate} = k[A]^2[B]$

d. $\text{Rate} = k[B]$

DR MOHAMED ABDELSALAM 0502500589

Given the experimental data in the table below and using the method of initial rates.

What is the rate law for the following reaction?

(Hint: Any number to the zero power equals one)

بمعرفة البيانات التجريبية في الجدول أدناه ومستخدمًا طريقة السرعات الابتدائية.

ما قانون سرعة التفاعل التالي؟

(تلميح: أي عدد يكون الأس الخاص به صفرًا يساوي 1)

$aA + bB \rightarrow products$ (نواتج)			
Experimental Data البيانات التجريبية			
السرعة الابتدائية Initial Rate mol/(L.s)	التركيز الابتدائي Initial concentration $[B]$ (M)	التركيز الابتدائي Initial concentration $[A]$ (M)	تجربة Trial
2.00×10^{-3}	0.100	0.100	1
2.00×10^{-3}	0.100	0.200	2
4.00×10^{-3}	0.200	0.200	3

a. Rate = k [A] [B]

b. Rate = k $[A]^2$ [B]

c. Rate = k [A]

d. Rate = k [B]

DR MOHAMED ABDELSALAM 0502500589

Using the experimental data in the following table
What is the rate law for the reaction?

مستخدما البيانات التجريبية في الجدول التالي ، ما قانون السرعة للتفاعل؟

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

التجربة Trial	التركيز الابتدائي Initial concentration [NO] (M)	التركيز الابتدائي Initial concentration $[O_2]$ (M)	السرعة الابتدائية Initial Rate (mol/(L.s))
1	0.030	0.020	0.0041
2	0.060	0.020	0.0164
3	0.060	0.040	0.0328

a. $\text{Rate} = k[NO][O_2]$

b. $\text{Rate} = k[NO]^2[O_2]^2$

c. $\text{Rate} = k[NO][O_2]^2$

d. $\text{Rate} = k[NO]^2[O_2]$

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21	CHM.5.4.01.005.02 Explain the function of the specific rate constant (k) in rate law equation	Textbook

The General Rate Law

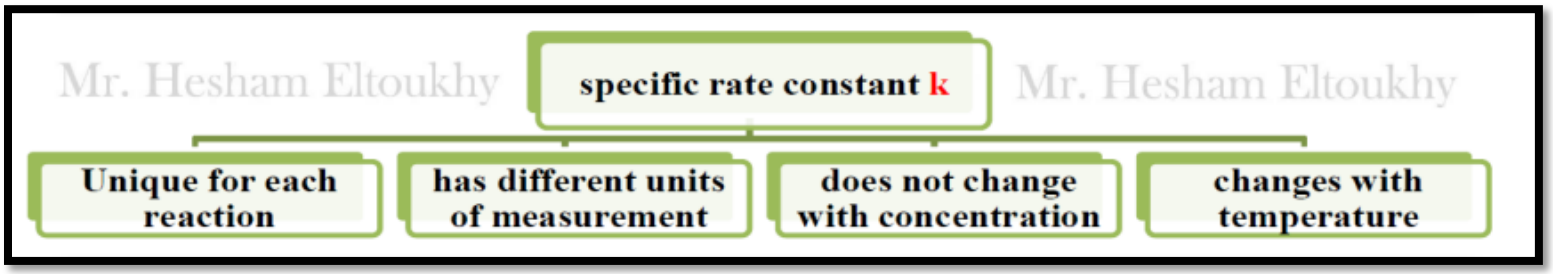
$$\text{rate} = k[\text{A}]^m[\text{B}]^n$$

[A] and [B] represent the concentrations of reactants A and B. The exponents *m* and *n* are the reaction orders.

The rate of a reaction is equal to the product of *k* and the concentrations of the reactants each raised to a power (order) that is determined experimentally.

The symbol *k* is the **specific rate constant,** a numerical value that relates the reaction rate and the concentrations of reactants at a given temperature. The specific rate constant is unique for every reaction and can have a variety of units including L/(mol•s), $L^2/(mol^2 \cdot s)$, and s^{-1}. A rate law must be determined experimentally as illustrated in **Figure 15** on the previous page.

The rate law shows that the reaction rate is directly proportional to the molar concentration of A. The specific rate constant, *k*, does not change with concentration; however, *k* does change with temperature. A large value of *k* means that A reacts rapidly to form B.



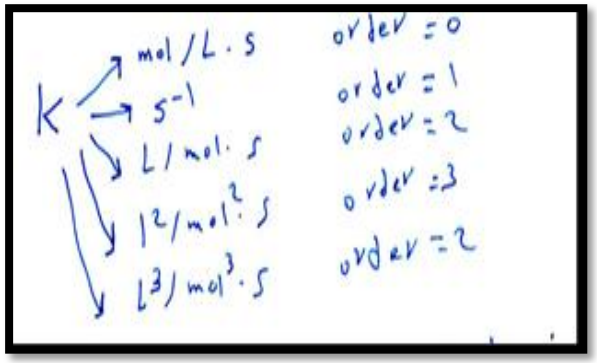
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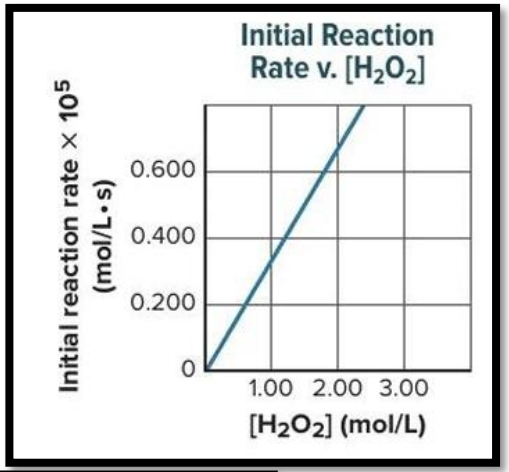
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$$\text{Rate} = [\text{reactant}]^{\text{order}}$$

For a zero-order reactant, changing its concentration does not affect the rate of the reaction, which means the rate does not depend on the concentration of this reactant.

reaction order	1	2	3	4
k unit	s^{-1}	$M^{-1}s^{-1}$	$M^{-2}s^{-1}$	$M^{-3}s^{-1}$
	s^{-1}	$L/(mol.s)$	$L^2/(mol^2.s)$	$L^3/(mol^3.s)$
rate unit	mol/(L.s)			





The rate law for a certain reaction:

قانون السرعة لتفاعل ما:

is $\text{Rate} = k[A][B]^2$

هو $\text{Rate} = k[A][B]^2$

if the concentration of **B** is **halved**, and all other conditions remain unchanged, how does the reaction rate change?

إذا قل تركيز **B** إلى **النصف** وبقيت كل الظروف الأخرى ثابتة ، كيف تتغير سرعة التفاعل ؟

- Decreases to a quarter of the initial reaction rate value — تقل إلى ربع قيمة السرعة الابتدائية
- Quadruples (increases 4 times) — تتضاعف أربع مرات
- Decreases to half of the initial reaction rate value — تقل إلى نصف قيمة السرعة الابتدائية
- Doubles — تتضاعف

The following decomposition reaction:

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

is first order, and the graph below shows relationship between its rate of decomposition and its concentration, when $[H_2O_2]$ equals 1.50 mol/L

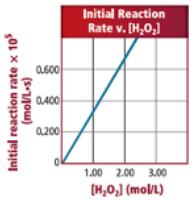
Which of the following is the **correct unit** of the specific rate constant **k**?

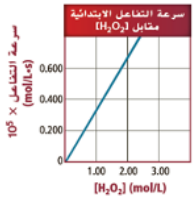
تفاعل التفكك التالي:

$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

من الرتبة الأولى، ويوضح الرسم البياني أدناه العلاقة بين سرعة تفككه وتركيزه، عندما يكون $[H_2O_2]$ يساوي 1.50 mol/L

أي مما يلي هو **الوحدة الصحيحة** لثابت السرعة النوعية **k**؟





a. $L/(mol.s)$

b. $L^2/(mol^2.s)$

c. s^{-1}

What is the unit of the specific rate constant k for a reaction that has the rate law:

Rate = $[A]^2[B]$?

a. s^{-1}

b. $L^2/(mol^2.s)$

c. $L^3/(mol^3.s)$

d. $L/(mol.s)$

Mr. Hesham Eltoukhy Mr. Hesham Eltoukhy Mr. Hesham Eltoukhy

DR MOHAMED ABDELSALAM 0502500589

The chemical reaction : التفاعل الكيميائي:

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

has the following rate law $Rate = k[NO]^2[O_2]$ if the concentration of NO decreases by half, what happens to the reaction rate?

له قانون السرعة التالي $Rate = k[NO]^2[O_2]$
إذا نقص تركيز NO إلى النصف، ماذا يحدث لسرعة التفاعل ؟

a. Reduces to one-half — تنقص إلى النصف

b. Becomes double — تُصبح الضعف

c. Reduces to one-quarter — تنقص إلى الربع

d. Remains the same — تبقى كما هو

Assume that a general chemical reaction has a rate law of Rate = $[A]^2[B]^3$ and the rate of the reaction under known conditions is 4.5 x 10^{-4} mol/(L.min) if the concentrations of A and B are doubled and all other reaction conditions remain constant. How will the reaction rate change?

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a. the reaction speed increases by a factor of 4 to become 1.8 x 10^{-3} mol/(L.min)

b. the reaction speed increases by a factor of 8 to become 3.6 x 10^{-3} mol/(L.min)

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c. the reaction speed increases by a factor of 16 to become 7.2 x 10^{-3} mol/(L.min)

d. the reaction speed increases by a factor of 32 to become 1.44 x 10^{-2} mol/(L.min)

The rate law for the following reaction:

$$aA + bB \rightarrow cC + dD$$

is $\mathbf{Rate = k[A]^2[B]^1}$

What happens to the rate if the concentration of **A** doubles?

قانون السرعة للتفاعل التالي :

$$aA + bB \rightarrow cC + dD$$

هو $\mathbf{Rate = k[A]^2[B]^1}$

ماذا يحدث للسرعة إذا تضاعف تركيز **A**؟

a. The rate reduces by half — تقل السرعة بمقدار النصف

b. The rate increases four times — تزداد السرعة أربعة أضعاف

c. The rate increases three times — تزداد السرعة ثلاثة أضعاف

d. The rate doubles — تتضاعف السرعة

DR MOHAMED ABDELSALAM 0502500589

The reaction below is second order in O_2 and fourth order overall. What is the rate law for the reaction?

التفاعل أدناه من الرتبة الثانية في O_2 ومن الرتبة الرابعة من حيث الرتبة الكلية. ما قانون سرعة التفاعل؟

$$2SO_{2(g)} + O_{2(g)} \longrightarrow 2SO_{3(g)}$$

a. $\text{Rate} = k\,[SO_2]^3\,[O_2]$

b. $\text{Rate} = k\,[O_2]^4\,[SO_2]$

c. $\text{Rate} = k\,[O_2]^2\,[SO_2]^2$

d. $\text{Rate} = k\,[SO_2]\,[O_2]^3$

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22	CHM.5.4.01.020.08 Calculate the instantaneous rate of a reaction from experimental data - calculate the concentration of a reactant using the instantaneous rate of a reaction	Textbook + Example problem 2 + Practice problems
23	CHM.5.4.01.020.08 Calculate the instantaneous rate of a reaction from experimental data	Textbook + Example problem 2 + Practice problems

The rate law for the reaction: $A + B + C \rightarrow products$ is :

$$Rate = k[B]^2[C]$$

If $k=6.92\times10^{-5}$ $L^2/(mol^2.s)$

and [A]= 0.175 M , [B]= 0.230 M ,[C]= 0.315 M.

What is the instantaneous reaction rate?

قانون السرعة للتفاعل : $A + B + C \rightarrow$ نواتج هو:

$$Rate = k[B]^2[C]$$

إذا كان $k=6.92\times10^{-5}$ $L^2/(mol^2.s)$

و [A] =0.175 M، [B]=0.230 M، [C]=0.315 M

ما هي سرعة التفاعل اللحظية؟

- 1.15×10^{-6} mol/(L.s)
- 8.77×10^{-7} mol/(L.s)
- 6.41×10^{-7} mol/(L.s)
- 5.01×10^{-6} mol/(L.s)

CALCULATE INSTANTANEOUS REACTION RATES The following reaction is first order in H_2 and second order in NO with a rate constant of 2.90×10^2 ($L^2/(mol^2 \cdot s)$).

$$2NO(g) + H_2(g) \rightarrow N_2O(g) + H_2O(g)$$

Calculate the instantaneous rate when the reactant concentrations are $[NO] = 0.00200M$ and $[H_2] = 0.00400M$.

Challenge Calculate [NO] for the reaction in Example Problem 2 if the rate is 9.00×10^{-5} mol/(L•s) and $[H_2]$ is $0.00300M$.

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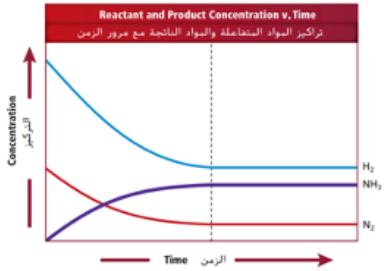
24	CHM.5.4.02.001.05 Describe chemical equilibrium using reactant and product concentration-time graph or particulate diagrams	Textbook

The graph below shows the change in the concentrations of reactants, and products during the reaction of nitrogen and hydrogen to form ammonia.

Which of the following is correct?

يُظهر الرسم البياني أدناه تغير تراكيز المواد المتفاعلة والناتجة خلال تفاعل النيتروجين والهيدروجين لإنتاج الأمونيا.

أي مما يأتي صحيح؟



Learning Outcomes Covered

- CHM.5.4.02.001

a. The concentrations of the reactants increase at first — تتزايد تراكيز المواد المتفاعلة في البداية

b. The concentrations of the reactants decrease at first — تتناقص تراكيز المواد المتفاعلة في البداية

c. At equilibrium, the concentrations of products increase — عند الاتزان تتزايد تراكيز المواد الناتجة

d. At equilibrium, the concentrations of products decrease — عند الاتزان تتناقص تراكيز المواد الناتجة

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25	CHM.5.4.02.005.02 Write chemical equilibrium expression for a heterogeneous equilibrium system (Keq)	Textbook + Example problem 2 + Practice problems

Q.4: كتابة تعابير الاتزان في الأنظمة المتزنة Equilibrium expression's writing for systems that are at equilibrium **Mark(s): 5/5**

$\frac{[\quad][\quad]}{[HI]^2}$	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	A
$\frac{[\ Na_2CO_3][CO_2][H_2O]}{[NaHCO_3]^2}$	$2NaHCO_3(s) \rightleftharpoons Na_2CO_3(s) + CO_2(g) + H_2O(g)$	B
$\frac{[CH_4][H_2O]}{[CO][H_2]^3}$	$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$	C
$\frac{[CaCO_3]}{[CaO]}$	$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	D

a. A

b. B

c. C

d. D

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What is the equilibrium constant expression for the following reaction?

ما تعبير ثابت الاتزان للتفاعل التالي؟

$$C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)}$$

Learning Outcomes Covered

- CHM.5.4.02.005

a. $\frac{[CO\,][H_2\,]}{[C\,][H_2O\,]}$

b. $\frac{[CO\,][H_2\,]}{[H_2O\,]}$

c. $\frac{[C\,][\,H_2O\,\,]}{[CO\,][\,H_2\,]}$

d. $\frac{[\,H_2O\,\,]}{[CO\,][\,H_2\,]}$

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26	CHM.5.4.02.006.02 Calculate the equilibrium concentration of a reactant given the concentration of other reactants, products and equilibrium constant at a certain temperature	Textbook + Example problem 3 + Practice problems

The reaction $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g}$ reaches equilibrium. If the equilibrium concentrations are:

$[SO_2] = 0.0170$ mol / L

$[O_2] = 0.0230$ mol / L,

Keq is 4.32

What is the equilibrium concentration of SO_3 in mol / L

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO$ يصل إلى حالة

كانت التراكيز عند الاتزان هي:

$[SO_2] = 0.0170$ و $[O_2] = 0.0230$ mol / L

ن K_{eq} لهذا التفاعل هو 4.32

الاتزان لـ SO_3 بوحدة mol / L ؟

Learning Outcomes Covered

- CHM.5.4.02.001

a. 0.00536

b. 0.0411

c. 0.6850

d. 0.9970

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The reaction below reaches equilibrium at a certain temperature

يصل التفاعل أدناه إلى حالة الاتزان عند درجة حرارة معينة،

$$2NbCl_{4\,(g)} \rightleftharpoons NbCl_{3(g)} + NbCl_{5(g)}$$

, $K_{eq} = 6.90 \times 10^{-4}$, If the equilibrium concentrations are:

و $K_{eq} = 6.90 \times 10^{-4}$ ، إذا كانت تراكيز الاتزان هي:

$NbCl_3 = 0.450$ mol/L, $NbCl_5 = 0.0380$ mol/L

What is the equilibrium concentration of $NbCl_4$?

فما تركيز الاتزان لـ $NbCl_4$؟

Learning Outcomes Covered

- CHM.5.4.02.006
- CHM.5.4.02.010
- CHM.5.4.02.021
- CHM.5.5.02.006

4.98 mol/L

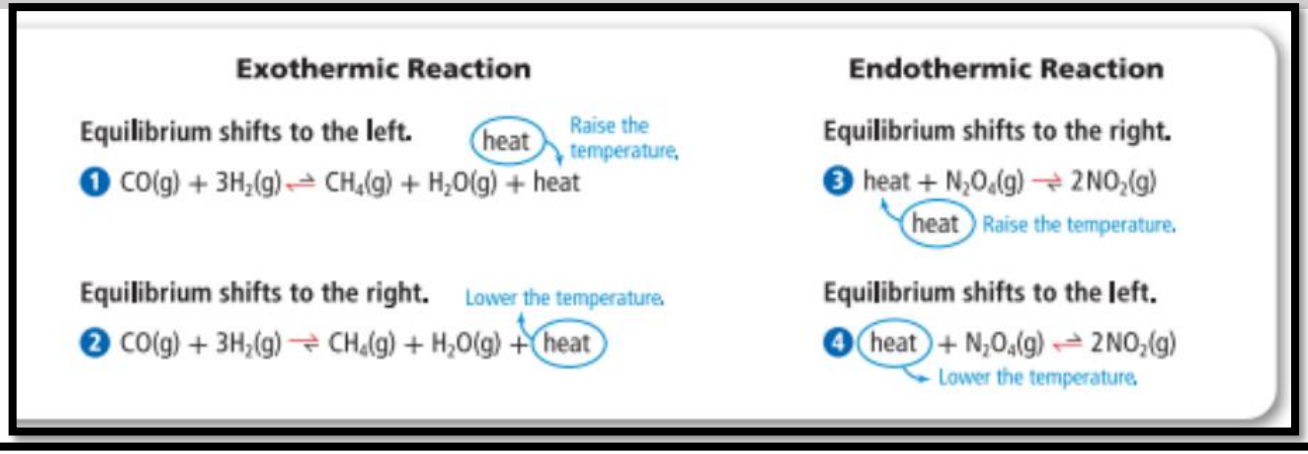
2.75 mol/L

1.69 mol/L

5.65 mol/L

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27	CHM.5.4.02.003.04 Explain the effect of changing temperature on an equilibrium system	Textbook + figure 14



The chemical equilibrium system used to produce **ethane C_2H_6** is: نظام الاتزان الكيميائي المستخدم في إنتاج **الإيثان C_2H_6** هو:

$$C_2H_4(g) + H_2(g) + (heat)\ \text{الحرارة} \rightleftharpoons C_2H_6(g)$$

Which of the following **increases** the **ethane C_2H_6** production? أي مما يلي **يزيد** من إنتاج **الإيثان C_2H_6**؟

إضافة C_2H_4 للنظام Adding C_2H_4 to the system	1
إزالة H_2 من النظام Removing H_2 from the system	2
زيادة درجة **الحرارة** Increasing temperature	3
تقليل الحجم Decreasing volume	4

- 1 ، 2 و4
- 2 ، 3 و4
- 1 ، 3 و4
- 1 ، 2 و3

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The reaction in the equation below is endothermic.

التفاعل في المعادلة أدناه ماص للحرارة.

Which of the following is correct?

أي مما يأتي صحيح؟

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \quad \Delta H^\circ = 57.2\ kJ$$

عديم اللون بني محمر



1 2

Learning Outcomes Covered

- CHM.5.4.02.006
- CHM.5.4.02.010
- CHM.5.4.02.021
- CHM.5.5.02.006

a. عند وضع وعاء التفاعل في حمام ساخن يظهر اللون 2

Placing reaction vessel in a boiling-water bath the color 2 appears

b. عند وضع وعاء التفاعل في الثلج يظهر اللون 2

Placing reaction vessel in ice the color 2 appears

c. عند وضع وعاء التفاعل في حمام ساخن ينزاح الاتزان جهة اليسار

Placing reaction vessel in a boiling-water bath the equilibrium shifts to the left

d. عند وضع وعاء التفاعل في الثلج ينزاح الاتزان جهة اليمين

Placing reaction vessel in ice the equilibrium shifts to the right

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28	CHM.5.4.02.003.03 Explain the effect of changing the volume and pressure on an equilibrium system	Textbook + fiqure 13

What will be the result if the volume of the reaction vessel is decreased for the reaction?

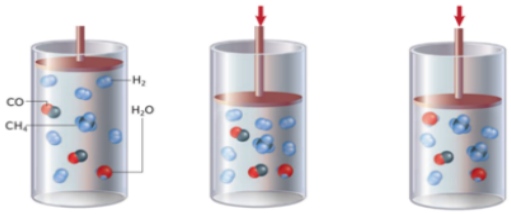
ما نتيجة تقليل حجم وعاء التفاعل التالي؟

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$$

The concentration of the product increaes	يزداد تركيز الناتج
The equilibrium shifts to the right	يتجه الاتزان نحو اليمين
The equilibrium does not change	لا يتغير الاتزان
The equilibrium shifts to the left	يتجه الاتزان نحو اليسار

When lowering the piston in the figure below, what happens to the next equilibrium reaction?

عند ضغط المكبس إلى أسفل في الشكل أدناه ماذا يحدث لتفاعل الاتزان التالي؟



$$CO_{(g)} + 3H_{2(g)} \rightleftharpoons CH_{4(g)} + H_2O_{(g)}$$

A –The equilibrium shifts to the left	A – ينزاح موضع الاتزان نحو اليسار
B – Increases the concentration of H_2	B– يزيد من تركيز H_2
C –The equilibrium shifts to the right	C – ينزاح موضع الاتزان نحو اليمين
D – Decreases the concentration of CH_4	D – يقل تركيز CH_4

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Regarding the equilibrium systems given in the table below.

Which of the following is correct?

فيما يتعلق بأنظمة الاتزان الواردة في الجدول أدناه.

أي مما يأتي صحيح؟

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$	1
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	2
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	3
$2N_2H_4(g) + 2NO_2(g) \rightleftharpoons 3N_2(g) + 4H_2O(g)$	4

Learning Outcomes Covered

- CHM.5.4.02.021

a. Decreasing the volume of the reaction vessel shifts the equilibrium **1** to the right

تقليل حجم وعاء التفاعل يُسبب إزاحة الاتزان **1** إلى اليمين

b. Decreasing the volume of the reaction vessel shifts the equilibrium **2** to the left

تقليل حجم وعاء التفاعل يسبب إزاحة الاتزان **2** إلى اليسار

c. Increasing the volume of the reaction vessel shifts the equilibrium **3** to the right

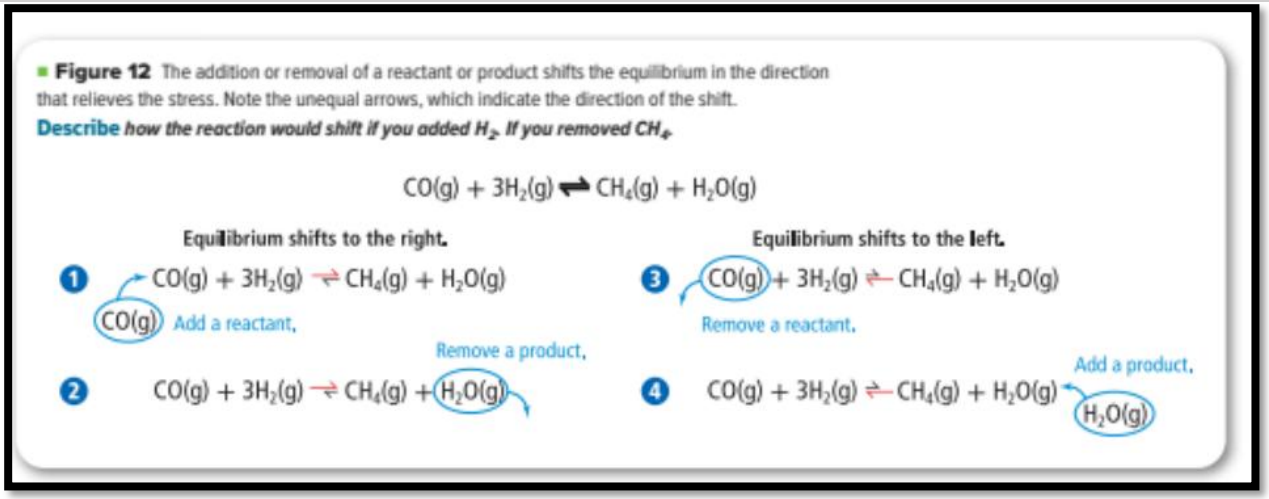
زيادة حجم وعاء التفاعل يسبب إزاحة الاتزان **3** إلى اليمين

d. Increasing the volume of the reaction vessel shifts the equilibrium **4** to the left

زيادة حجم وعاء التفاعل يسبب إزاحة الاتزان **4** إلى اليسار

D

29	CHM.5.4.02.003.02 Explain the effect of changing the concentration (Adding reactants or removing products or adding products) on an equilibrium system	Textbook + figure 12



All the following changes cause the equilibrium to shift to the left in the reaction below except .....

جميع التغيرات التالية تُسبب انزياح الاتزان إلى جهة اليسار في التفاعل أدناه عدا.....

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \quad \Delta H^\circ = 57.2\ kJ$$

Increasing the pressure	زيادة الضغط
Lowering the system temperature	خفض حرارة النظام
Reducing the concentration of NO_2	تقليل تركيز NO_2
Reducing the concentration of N_2O_4	تقليل تركيز N_2O_4

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Which of the following increases CH_3CHO production in the following equilibrium reaction?

أي مما يلي يؤدي إلى زيادة إنتاج CH_3CHO في تفاعل الاتزان التالية؟

$$C_2H_2(g) + H_2O(g) \rightleftharpoons CH_3CHO(g) \quad \Delta H^\circ = -151\ kJ$$

Decreasing temperature	خفض درجة الحرارة	1
Adding a lot of water	إضافة المزيد من الماء	2
Adding a catalyst	إضافة عامل حفاز	3
Adding a desiccant	إضافة عامل مجفف	4

a. 1 and 4 — 4 و1

b. 1 and 2 — 2 و1

c. 3 and 4 — 4 و3

d. 4 and 5 — 5 و4

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All of the following changes cause the equilibrium to shift to the left in the reaction below except .....

جميع التغيرات التالية تُسبب انزياح الاتزان إلى جهة اليسار في التفاعل أدناه عدا.....

$$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g) \quad \Delta H^\circ = -206.5\ kJ$$

- Raising the system temperature — رفع درجة حرارة النظام
- Adding a desiccant to the reaction vessel — إضافة عامل مجفف في وعاء التفاعل
- Decreasing the concentration of CO — تقليل تركيز CO
- Decreasing the pressure — تقليل الضغط

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30	CHM.5.4.02.003.06 Explain the effect of a catalyst on an equilibrium system	Textbook

Catalysts and equilibrium

Changes in concentration, volume, and temperature make a difference in the amount of product formed in a reaction. Can a catalyst also affect product concentration? A catalyst speeds up a reaction, but it does so equally in both directions. Therefore, a catalyzed reaction reaches equilibrium more quickly but with no change in the amount of product formed. To maximize efficiency, industrial chemists try to produce the most amount of product in the shortest amount of time possible. As a result, many industrial processes involve both finely tuned equilibrium conditions and catalysts.

Which of the following factors will NOT lead to an increase in the amount of substance produced?

ي العوامل التالية لن تؤدي إلى زيادة كمية المادة الناتجة؟

$$CO_{(g)} + 2H_2 \rightleftharpoons CH_3OH_{(g)}$$

Learning Outcomes Covered

- CHM.5.4.02.003
- CHM.5.4.02.006
- CHM.5.4.02.021

a. Adding **CO** to the system — إضافة CO للنظام

b. Removing **CH_3OH** from the system — إزالة CH_3O من النظام

c. Adding **H_2** to the system — إضافة H للنظام

d. Adding a catalyst to the system — إضافة حفاز للنظام

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What is the effect of a catalyst on a chemical reaction in equilibrium?

ما أثر الحفاز على تفاعل كيميائي في حالة اتزان؟

Learning Outcomes Covered

- CHM.5.4.02.021

a. Speeds up the forward reaction more than the reverse reaction

يزيد من سرعة التفاعل الأمامي أكثر من سرعة التفاعل العكسي

b. Speeds up the reverse reaction more than the forward reaction

يزيد من سرعة التفاعل العكسي أكثر من سرعة التفاعل الأمامي

c. Changes in the amount of product formed

يُغير في كمية المادة الناتجة المتكونة

d. The reaction reaches equilibrium more quickly

يصل التفاعل بشكل أسرع إلى حالة الاتزان

DR MOHAMED ABDELSALAM 0502500589